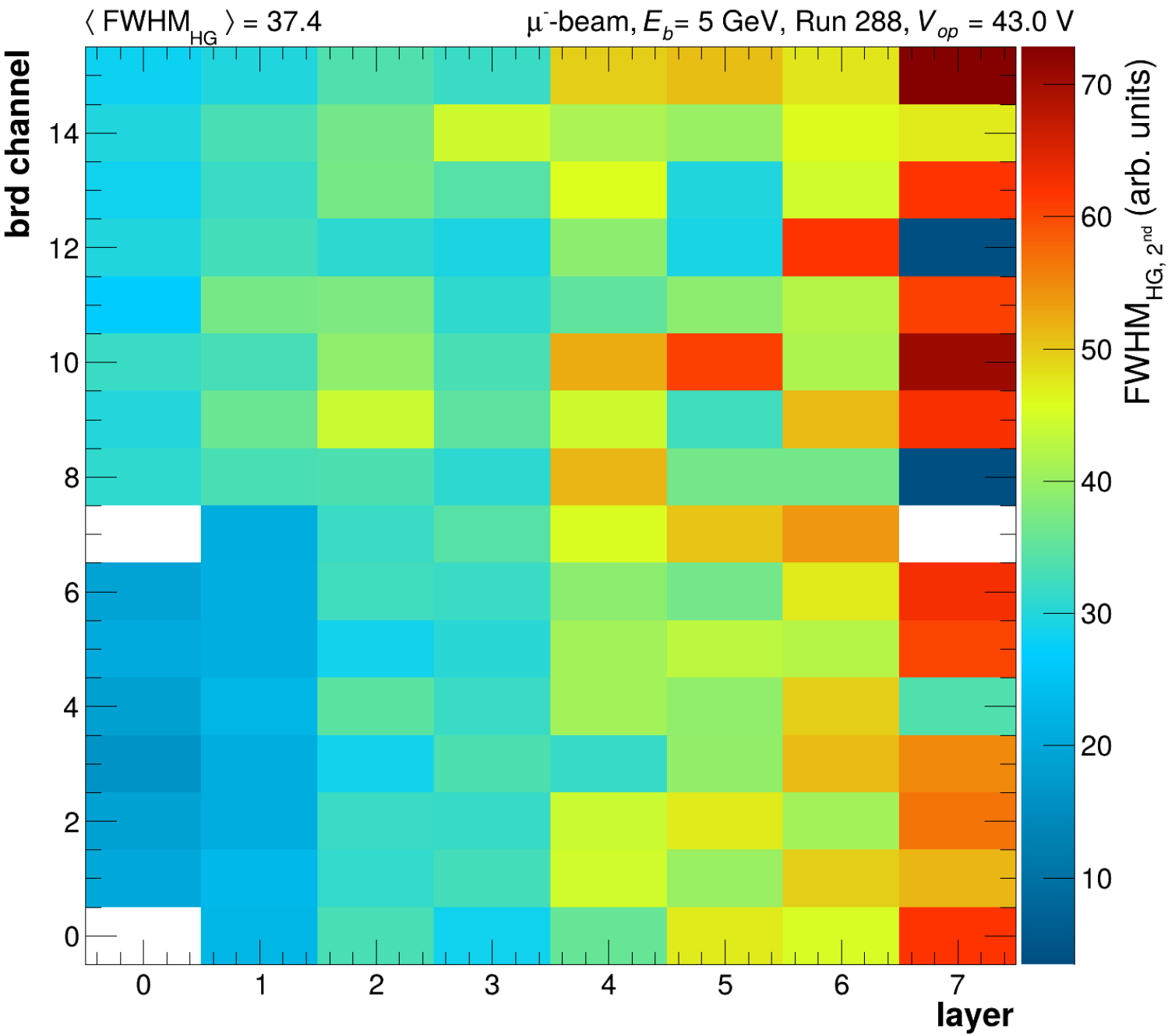
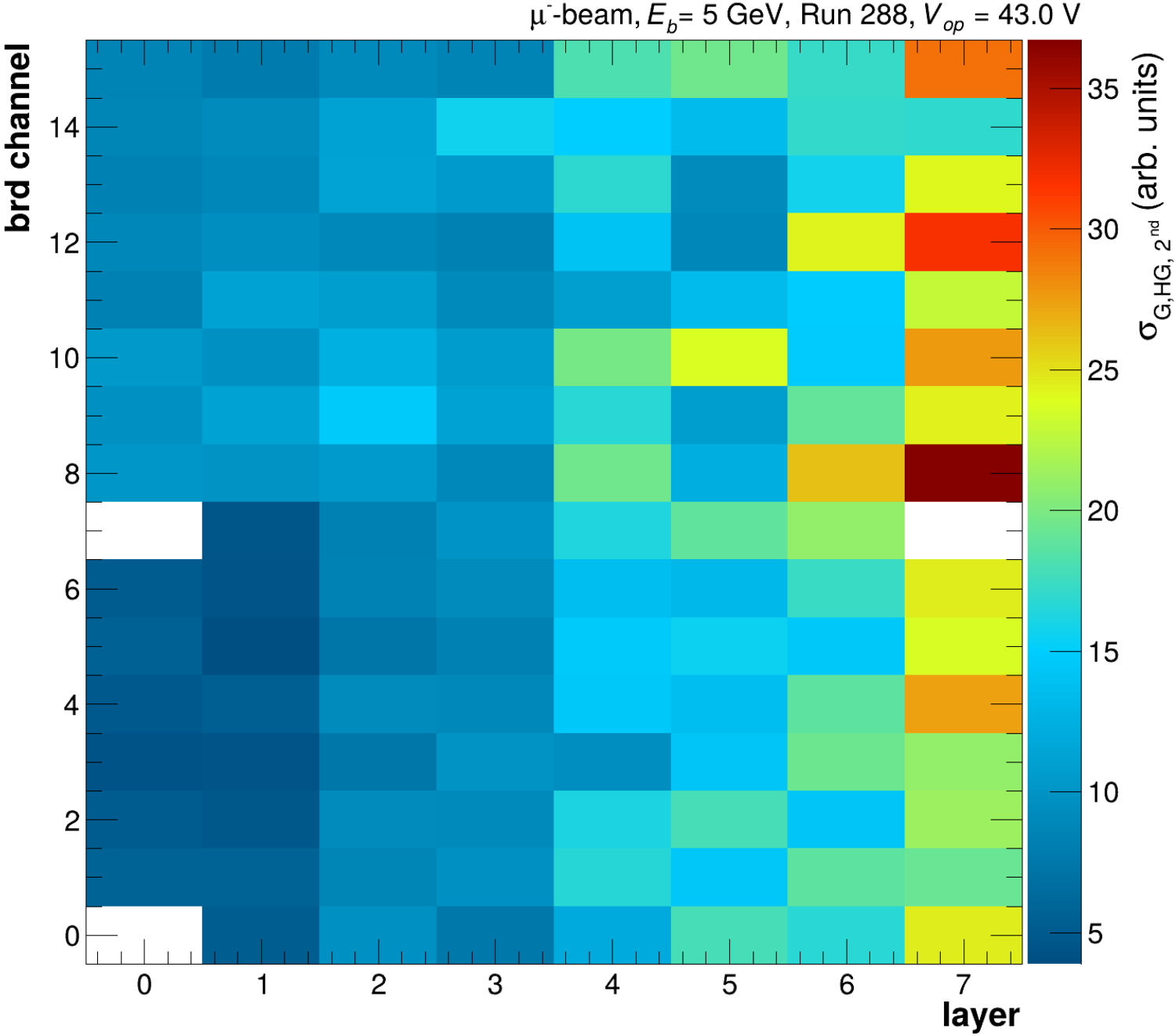
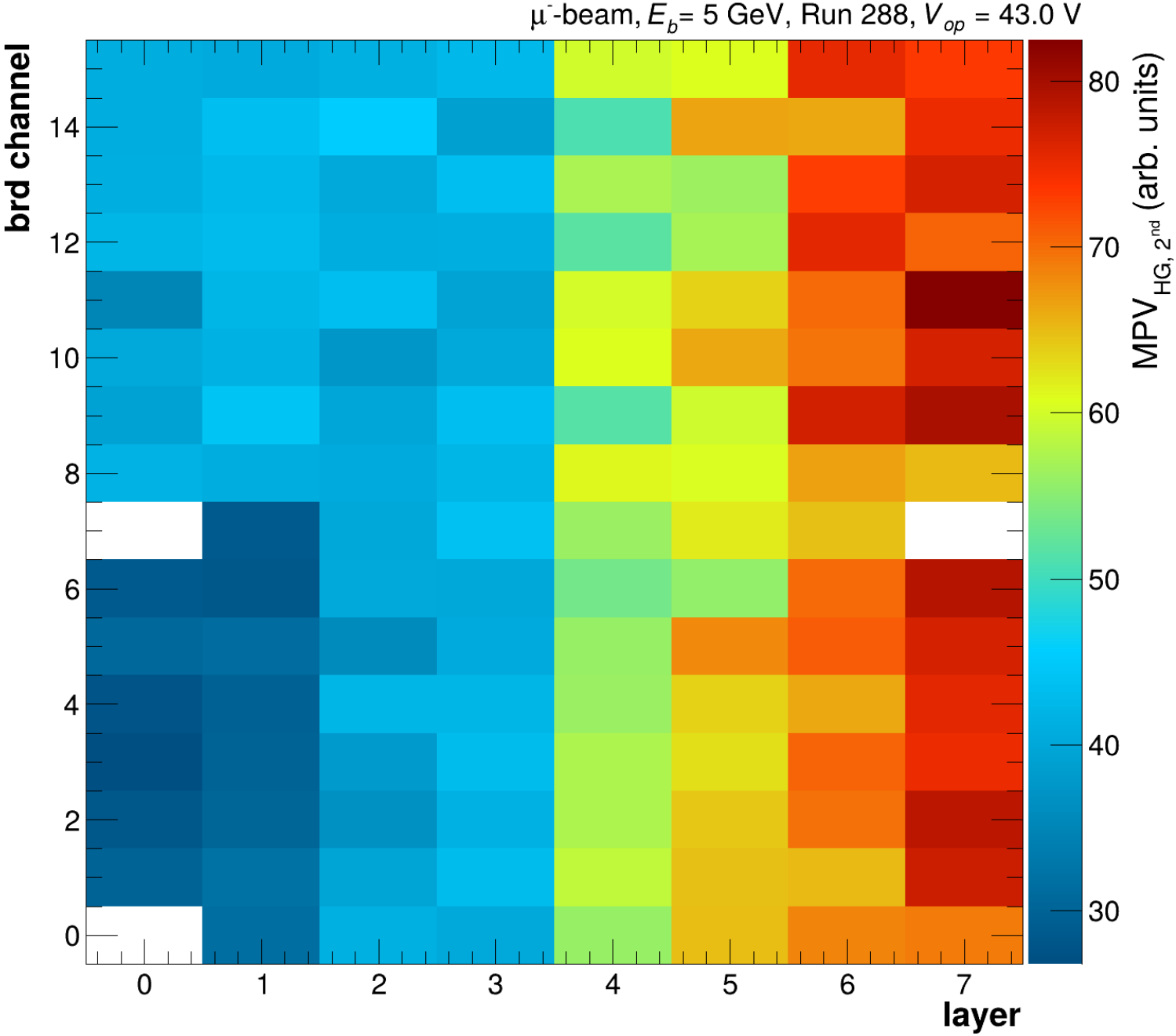
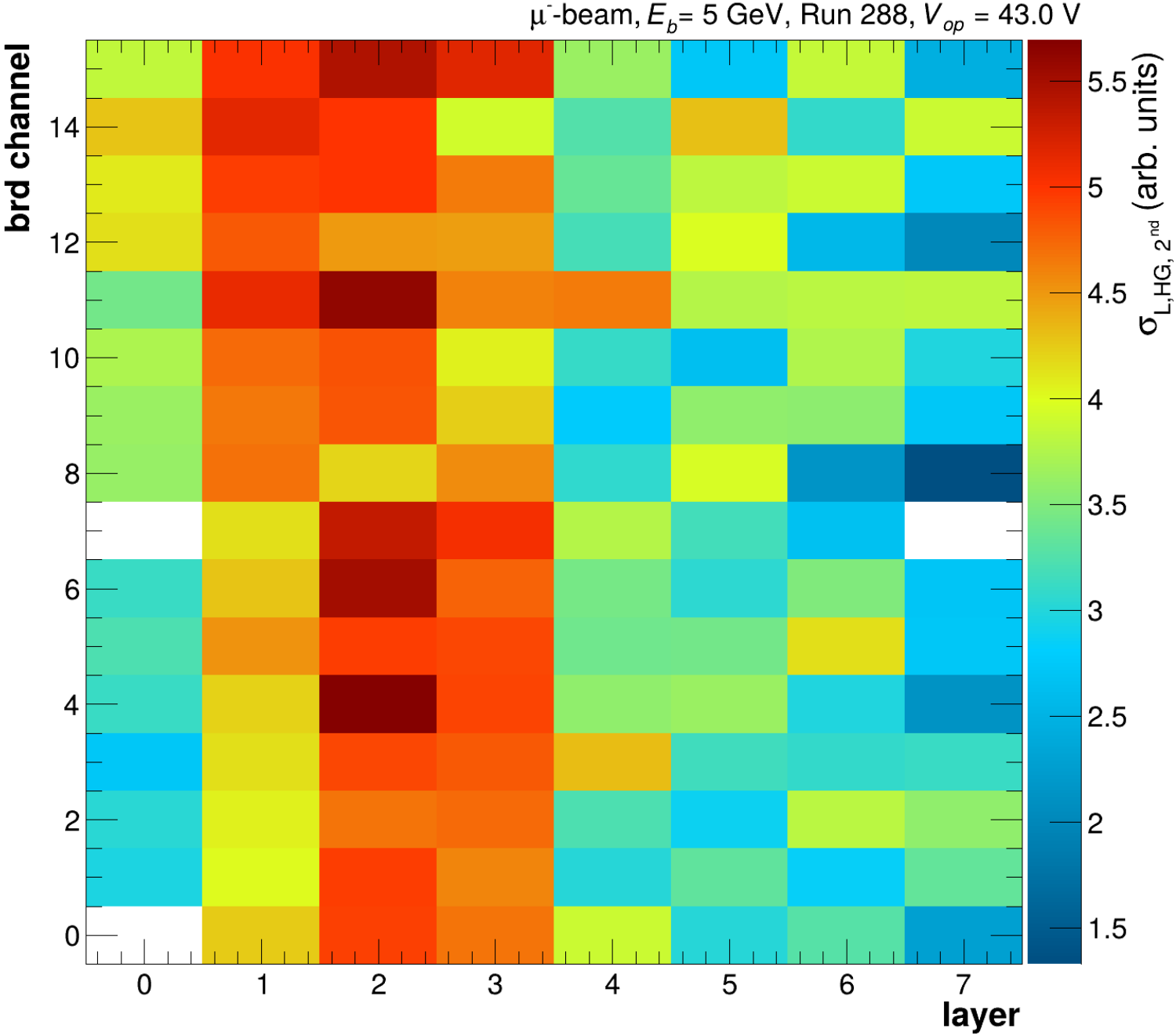
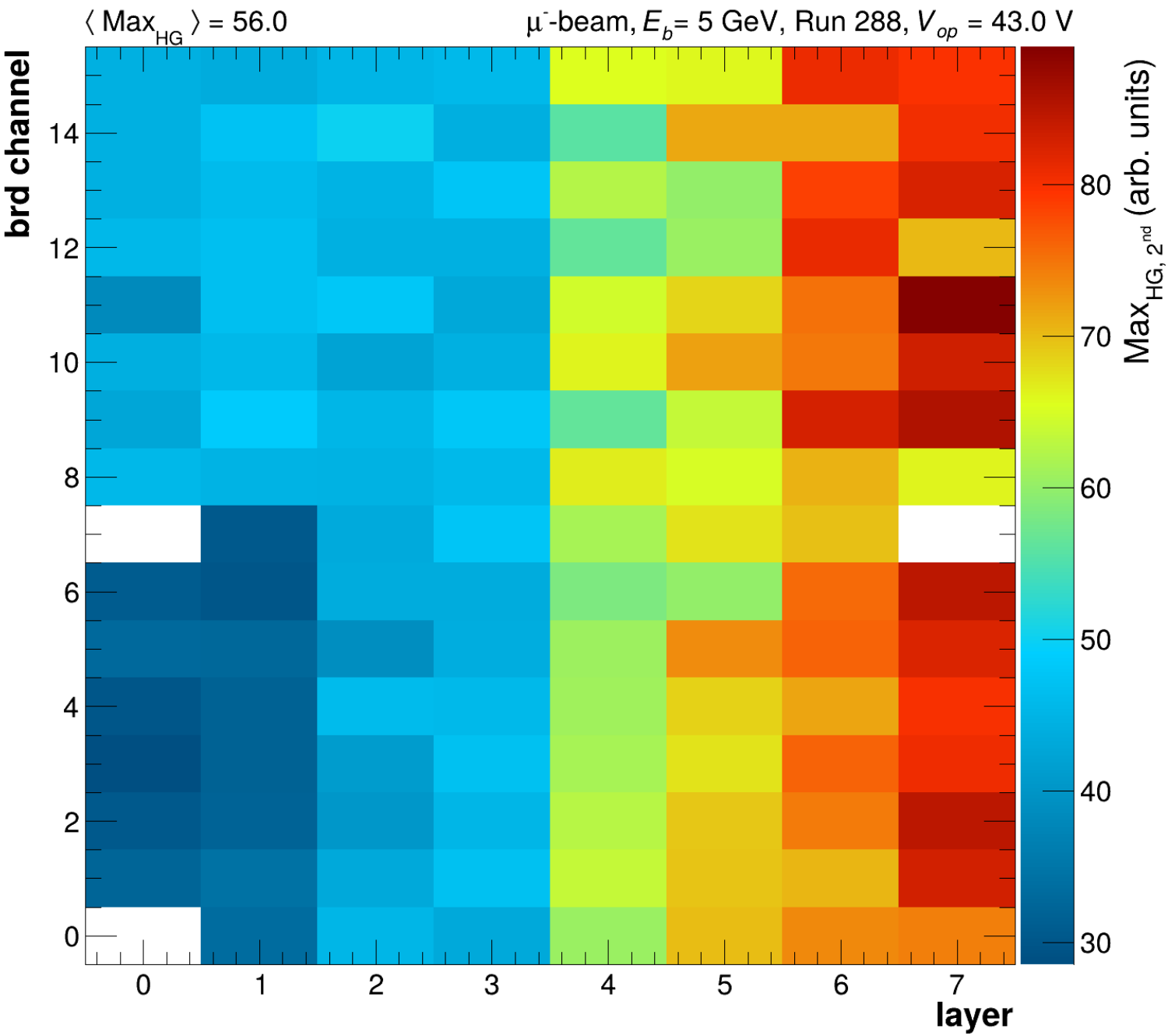
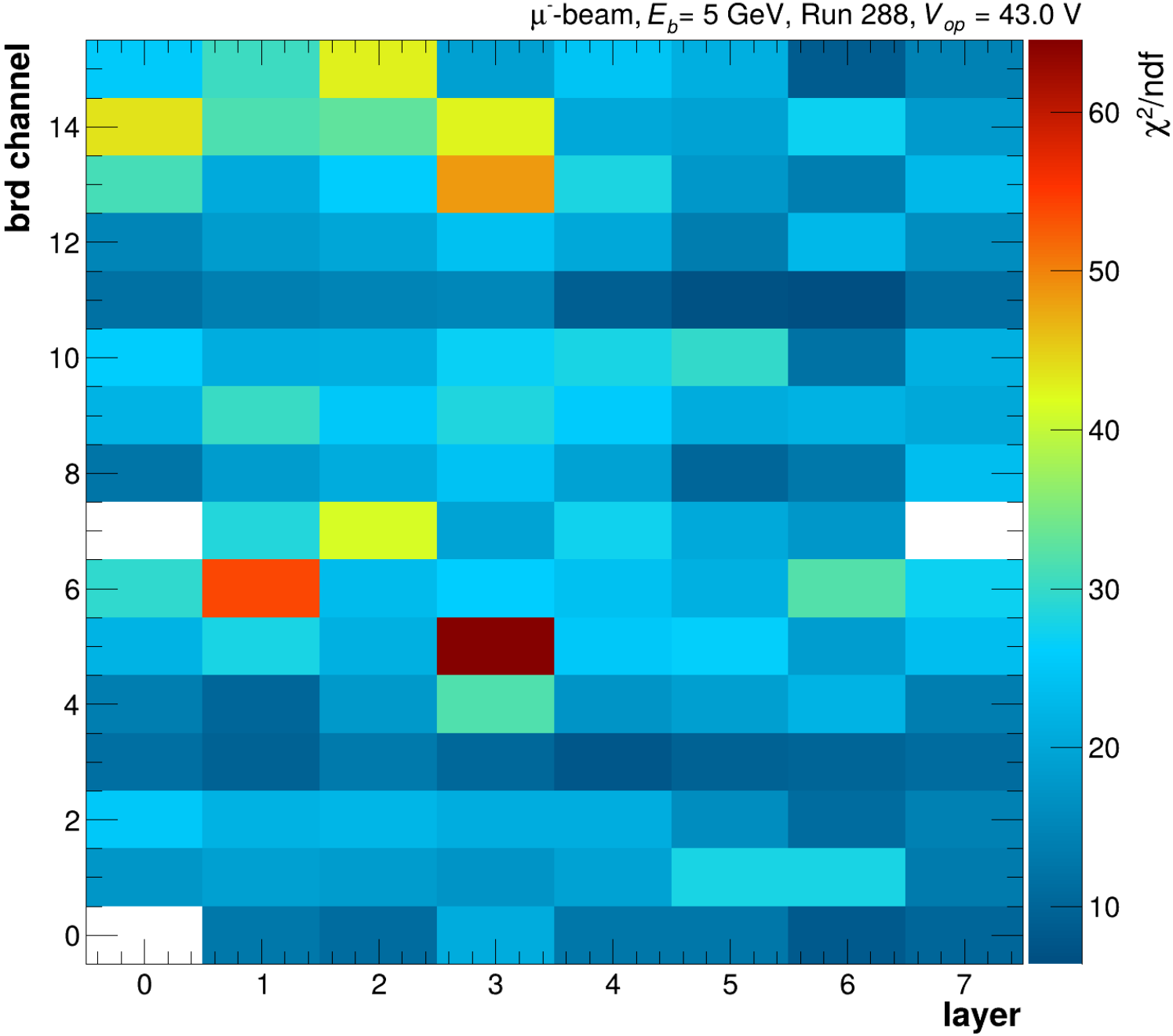

⟨ FWHM$_{HG}$ ⟩ = 37.4
μ⁻-beam, E_b= 5 GeV, Run 288, V_{op} = 43.0 V
brd channel
layer
FWHM$_{HG,\ 2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
1
3
5
7
10
20
30
40
50
60
70

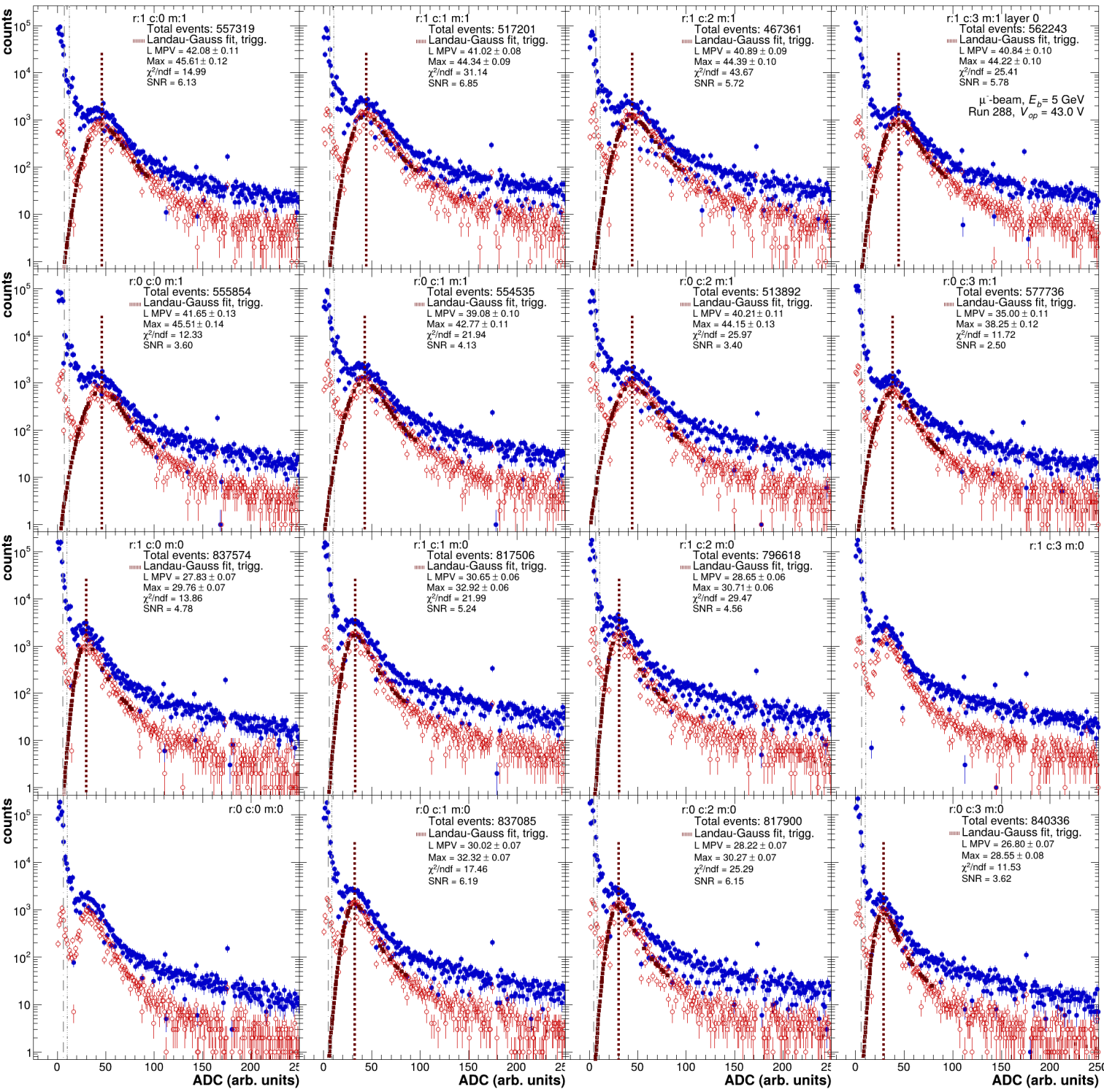

μ⁻-beam, E_b= 5 GeV, Run 288, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
$\sigma_{G,HG,2^{nd}}$ (arb. units)
5
10
15
20
25
30
35

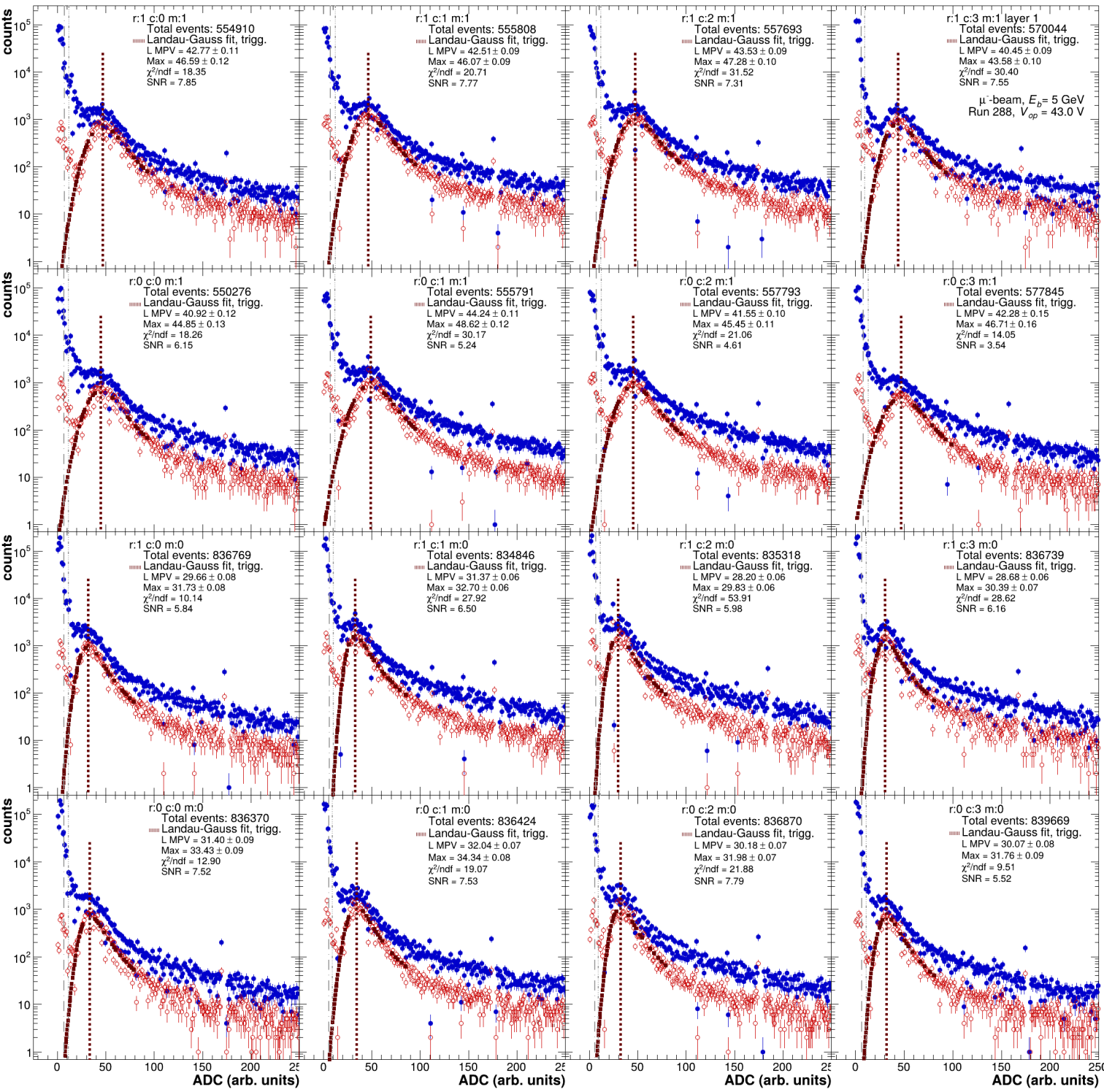

μ⁻-beam, E_b= 5 GeV, Run 288, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
MPV$_{HG,\,2^{nd}}$ (arb. units)
30
40
50
60
70
80

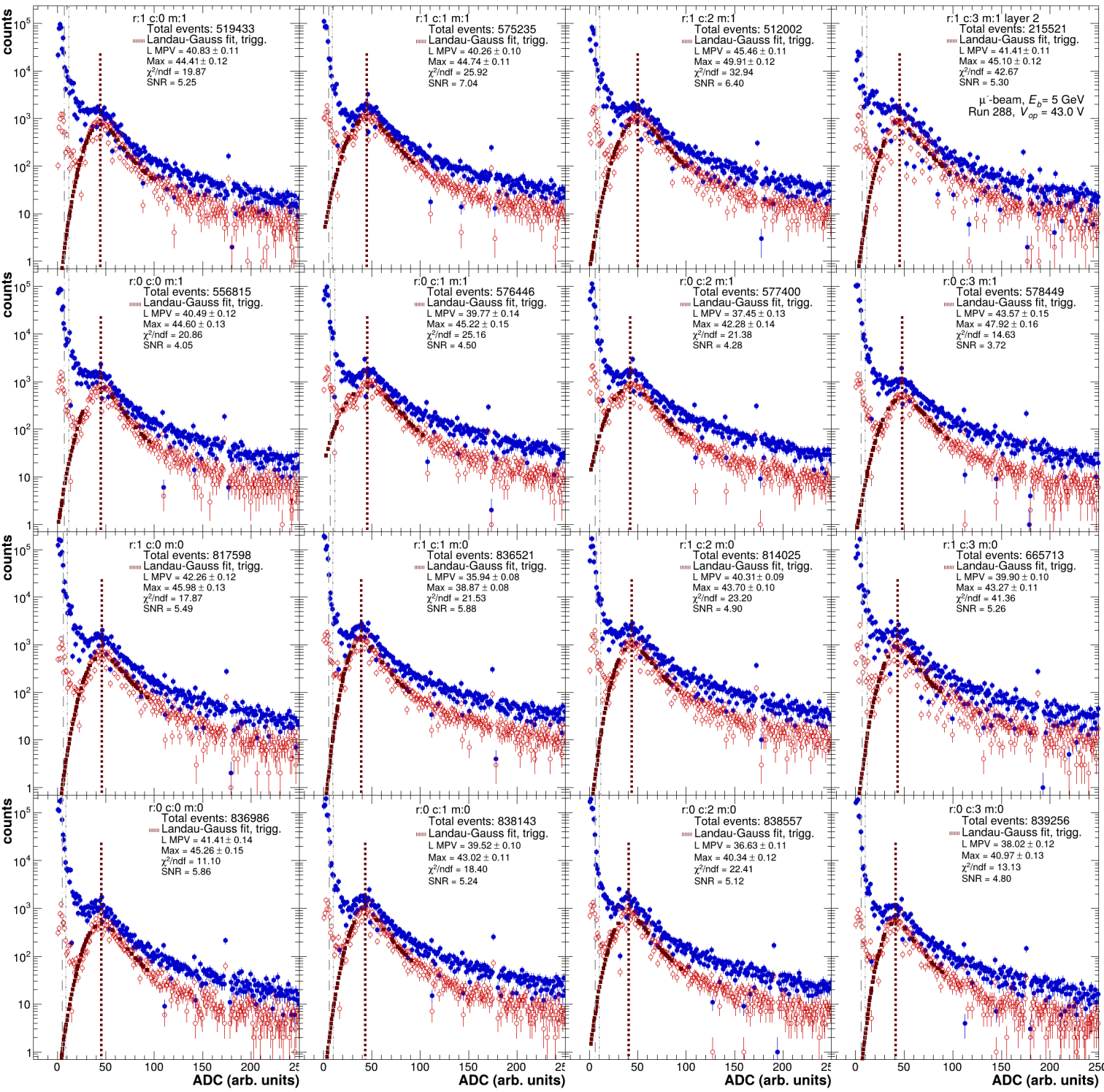

μ⁻-beam, E_b= 5 GeV, Run 288, V_{op} = 43.0 V
brd channel
layer
$\sigma_{L,HG,2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
1.5
2
2.5
3
3.5
4
4.5
5
5.5

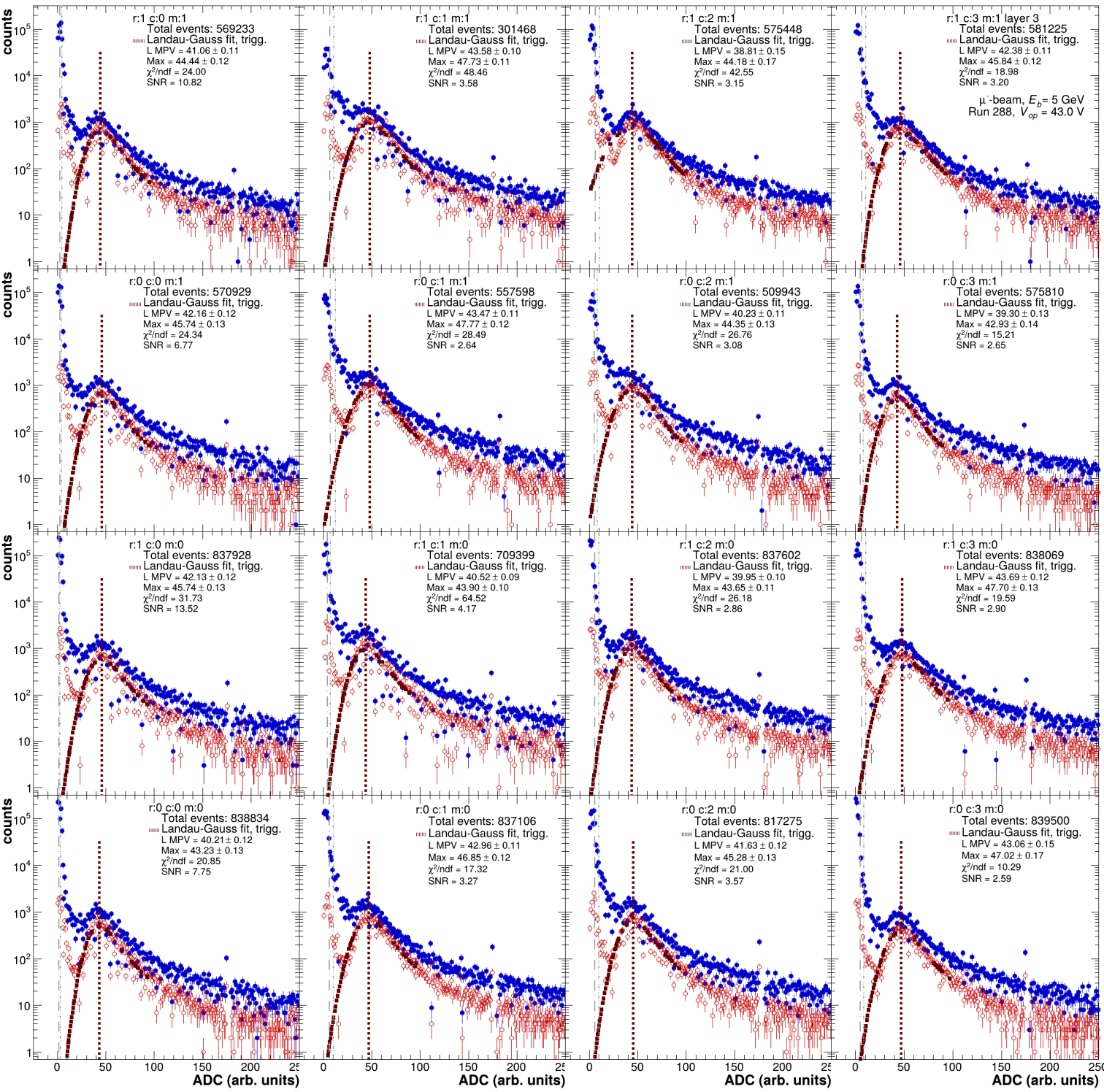

⟨ Max_HG ⟩ = 56.0
μ⁻-beam, E_b = 5 GeV, Run 288, V_op = 43.0 V
brd channel
layer
Max_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
30
40
50
60
70
80

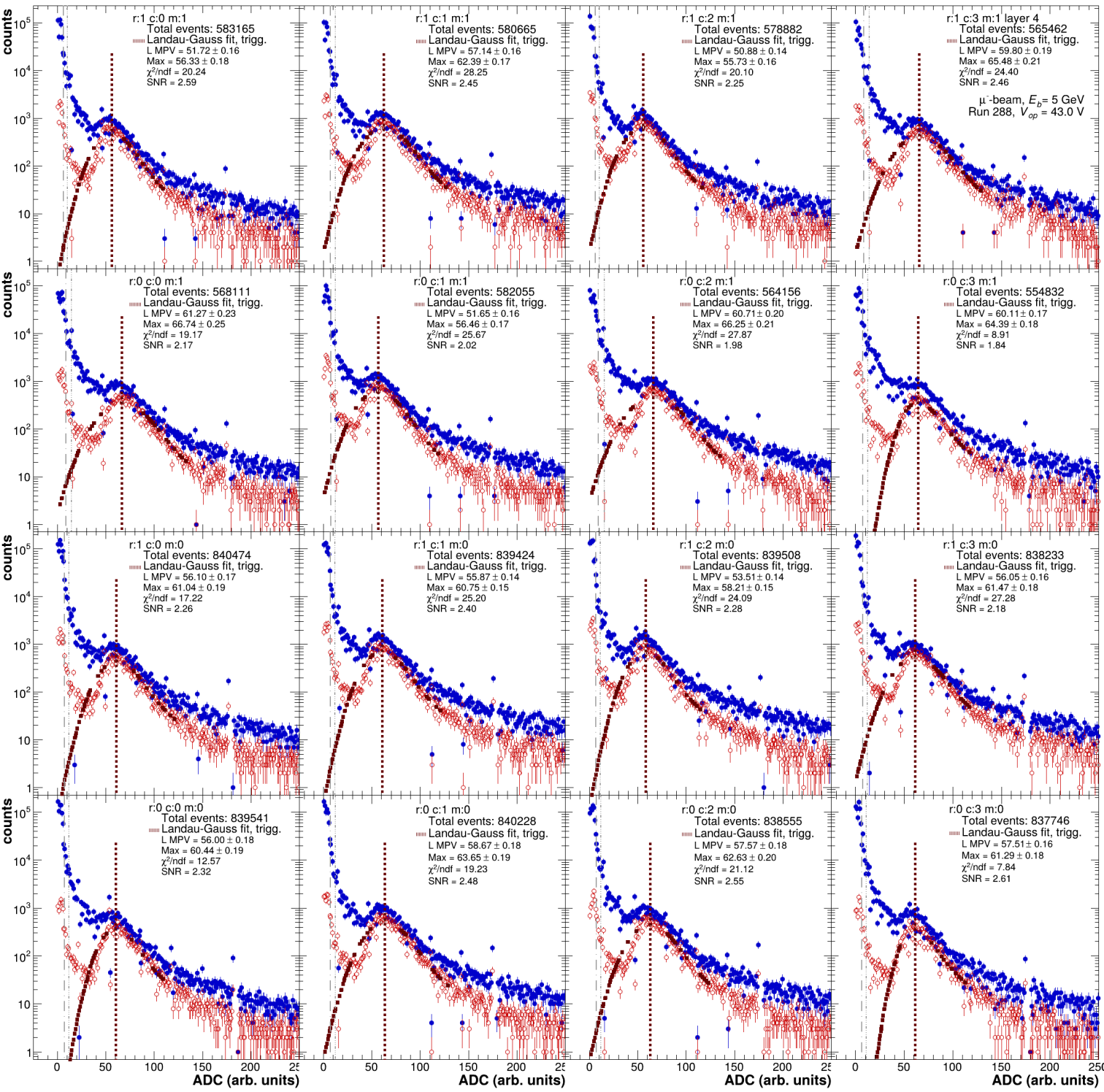

μ⁻-beam, E_b= 5 GeV, Run 288, V_{op} = 43.0 V
brd channel
layer
χ^2/ndf
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
10
20
30
40
50
60

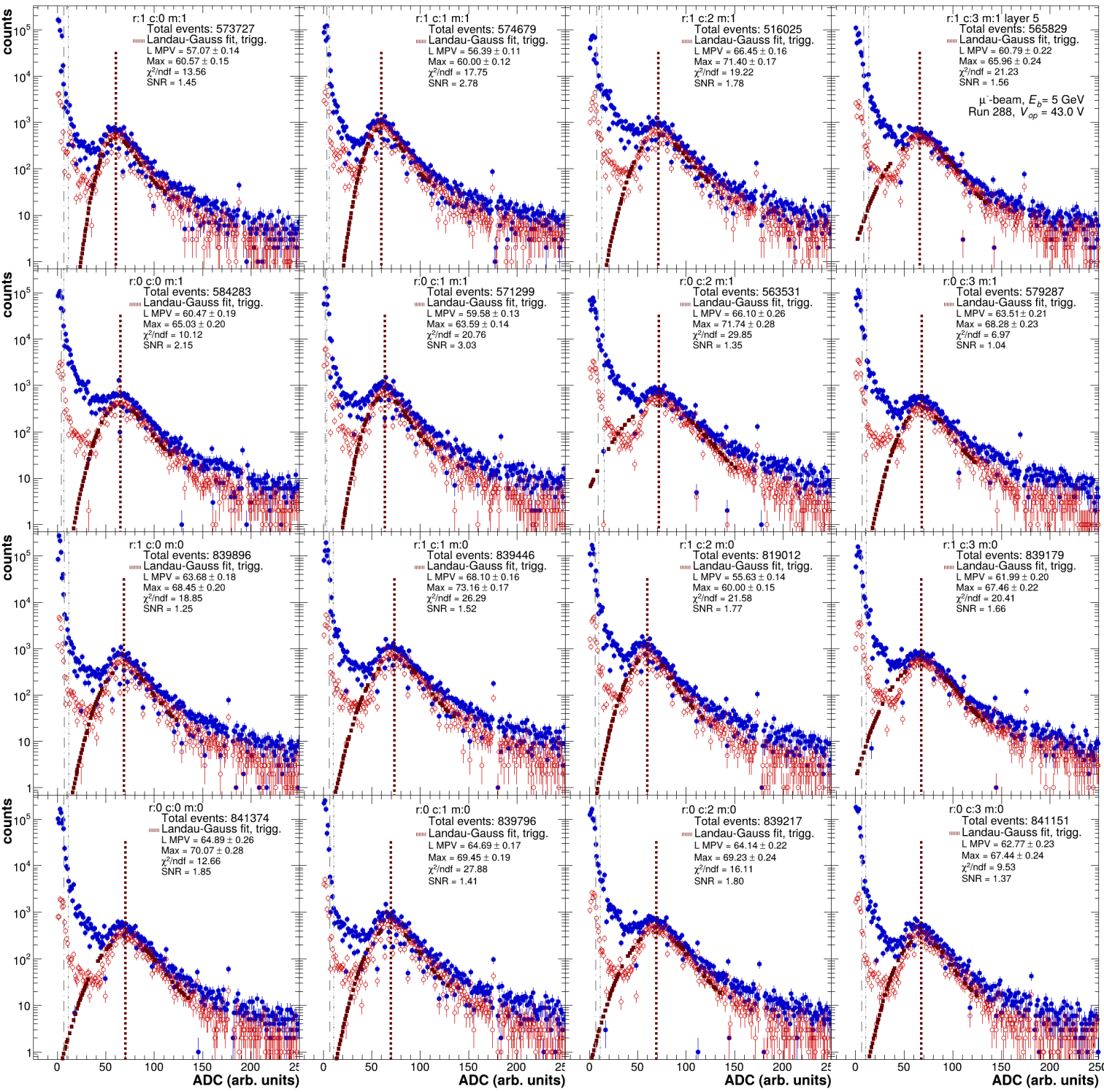

r:1 c:0 m:1
Total events: 557319
Landau-Gauss fit, trigg.
L MPV = 42.08 ± 0.11
Max = 45.61 ± 0.12
χ²/ndf = 14.99
SNR = 6.13
r:1 c:1 m:1
Total events: 517201
Landau-Gauss fit, trigg.
L MPV = 41.02 ± 0.08
Max = 44.34 ± 0.09
χ²/ndf = 31.14
SNR = 6.85
r:1 c:2 m:1
Total events: 467361
Landau-Gauss fit, trigg.
L MPV = 40.89 ± 0.09
Max = 44.39 ± 0.10
χ²/ndf = 43.67
SNR = 5.72
r:1 c:3 m:1 layer 0
Total events: 562243
Landau-Gauss fit, trigg.
L MPV = 40.84 ± 0.10
Max = 44.22 ± 0.10
χ²/ndf = 25.41
SNR = 5.78
μ⁻-beam, E_b = 5 GeV
Run 288, V_op = 43.0 V
r:0 c:0 m:1
Total events: 555854
Landau-Gauss fit, trigg.
L MPV = 41.65 ± 0.13
Max = 45.51 ± 0.14
χ²/ndf = 12.33
SNR = 3.60
r:0 c:1 m:1
Total events: 554535
Landau-Gauss fit, trigg.
L MPV = 39.08 ± 0.10
Max = 42.77 ± 0.11
χ²/ndf = 21.94
SNR = 4.13
r:0 c:2 m:1
Total events: 513892
Landau-Gauss fit, trigg.
L MPV = 40.21 ± 0.11
Max = 44.15 ± 0.13
χ²/ndf = 25.97
SNR = 3.40
r:0 c:3 m:1
Total events: 577736
Landau-Gauss fit, trigg.
L MPV = 35.00 ± 0.11
Max = 38.25 ± 0.12
χ²/ndf = 11.72
SNR = 2.50
r:1 c:0 m:0
Total events: 837574
Landau-Gauss fit, trigg.
L MPV = 27.83 ± 0.07
Max = 29.76 ± 0.07
χ²/ndf = 13.86
SNR = 4.78
r:1 c:1 m:0
Total events: 817506
Landau-Gauss fit, trigg.
L MPV = 30.65 ± 0.06
Max = 32.92 ± 0.06
χ²/ndf = 21.99
SNR = 5.24
r:1 c:2 m:0
Total events: 796618
Landau-Gauss fit, trigg.
L MPV = 28.65 ± 0.06
Max = 30.71 ± 0.06
χ²/ndf = 29.47
SNR = 4.56
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
Total events: 837085
Landau-Gauss fit, trigg.
L MPV = 30.02 ± 0.07
Max = 32.32 ± 0.07
χ²/ndf = 17.46
SNR = 6.19
r:0 c:2 m:0
Total events: 817900
Landau-Gauss fit, trigg.
L MPV = 28.22 ± 0.07
Max = 30.27 ± 0.07
χ²/ndf = 25.29
SNR = 6.15
r:0 c:3 m:0
Total events: 840336
Landau-Gauss fit, trigg.
L MPV = 26.80 ± 0.07
Max = 28.55 ± 0.08
χ²/ndf = 11.53
SNR = 3.62
counts
ADC (arb. units)

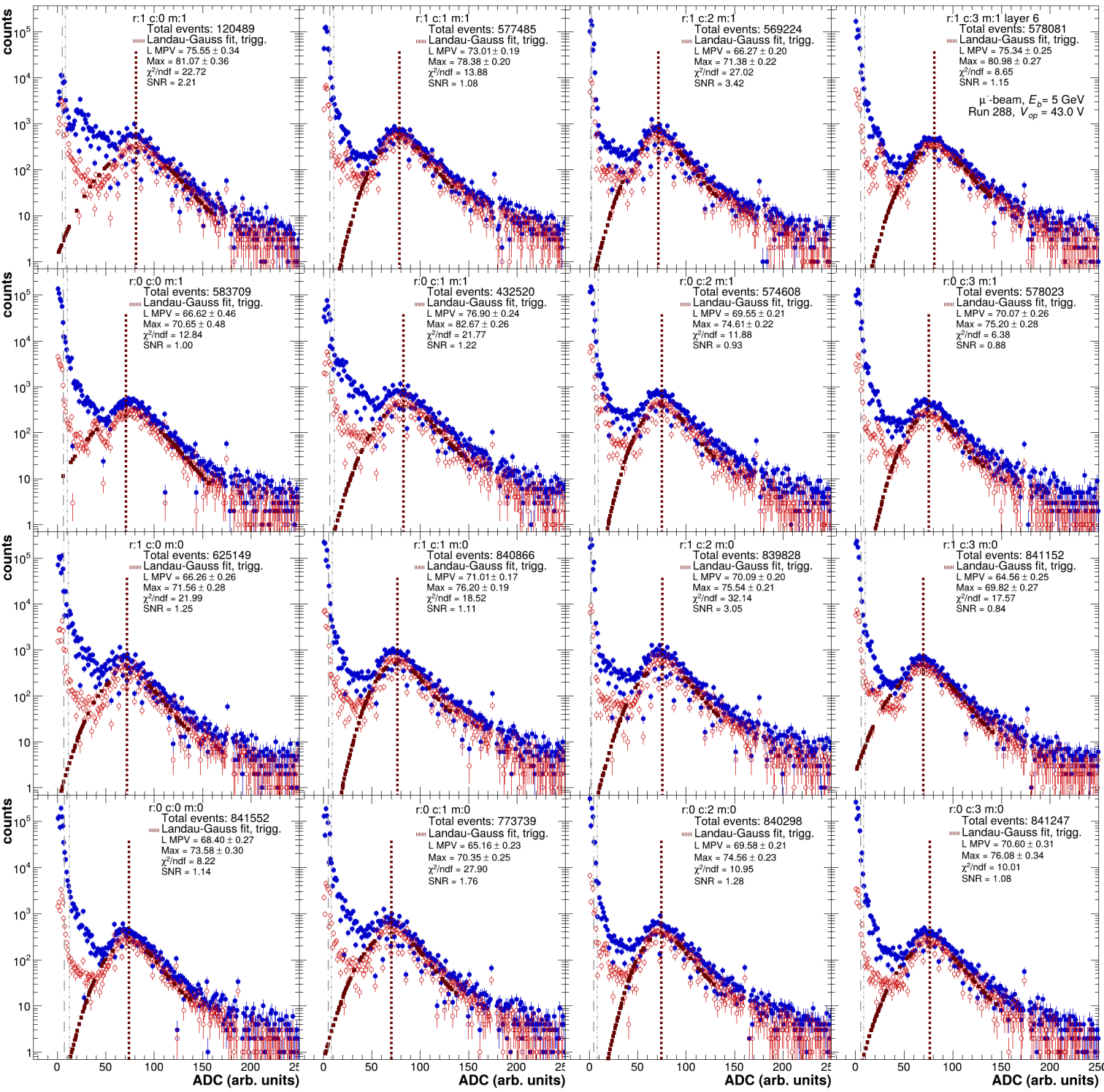

r:1 c:0 m:1
Total events: 554910
Landau-Gauss fit, trigg.
L MPV = 42.77 ± 0.11
Max = 46.59 ± 0.12
χ²/ndf = 18.35
SNR = 7.85
r:1 c:1 m:1
Total events: 555808
Landau-Gauss fit, trigg.
L MPV = 42.51 ± 0.09
Max = 46.07 ± 0.09
χ²/ndf = 20.71
SNR = 7.77
r:1 c:2 m:1
Total events: 557693
Landau-Gauss fit, trigg.
L MPV = 43.53 ± 0.09
Max = 47.28 ± 0.10
χ²/ndf = 31.52
SNR = 7.31
r:1 c:3 m:1 layer 1
Total events: 570044
Landau-Gauss fit, trigg.
L MPV = 40.45 ± 0.09
Max = 43.58 ± 0.10
χ²/ndf = 30.40
SNR = 7.55
μ⁻-beam, E_b = 5 GeV
Run 288, V_{op} = 43.0 V
r:0 c:0 m:1
Total events: 550276
Landau-Gauss fit, trigg.
L MPV = 40.92 ± 0.12
Max = 44.85 ± 0.13
χ²/ndf = 18.26
SNR = 6.15
r:0 c:1 m:1
Total events: 555791
Landau-Gauss fit, trigg.
L MPV = 44.24 ± 0.11
Max = 48.62 ± 0.12
χ²/ndf = 30.17
SNR = 5.24
r:0 c:2 m:1
Total events: 557793
Landau-Gauss fit, trigg.
L MPV = 41.55 ± 0.10
Max = 45.45 ± 0.11
χ²/ndf = 21.06
SNR = 4.61
r:0 c:3 m:1
Total events: 577845
Landau-Gauss fit, trigg.
L MPV = 42.28 ± 0.15
Max = 46.71 ± 0.16
χ²/ndf = 14.05
SNR = 3.54
r:1 c:0 m:0
Total events: 836769
Landau-Gauss fit, trigg.
L MPV = 29.66 ± 0.08
Max = 31.73 ± 0.08
χ²/ndf = 10.14
SNR = 5.84
r:1 c:1 m:0
Total events: 834846
Landau-Gauss fit, trigg.
L MPV = 31.37 ± 0.06
Max = 32.70 ± 0.06
χ²/ndf = 27.92
SNR = 6.50
r:1 c:2 m:0
Total events: 835318
Landau-Gauss fit, trigg.
L MPV = 28.20 ± 0.06
Max = 29.83 ± 0.06
χ²/ndf = 53.91
SNR = 5.98
r:1 c:3 m:0
Total events: 836739
Landau-Gauss fit, trigg.
L MPV = 28.68 ± 0.06
Max = 30.39 ± 0.07
χ²/ndf = 28.62
SNR = 6.16
r:0 c:0 m:0
Total events: 836370
Landau-Gauss fit, trigg.
L MPV = 31.40 ± 0.09
Max = 33.43 ± 0.09
χ²/ndf = 12.90
SNR = 7.52
r:0 c:1 m:0
Total events: 836424
Landau-Gauss fit, trigg.
L MPV = 32.04 ± 0.07
Max = 34.34 ± 0.08
χ²/ndf = 19.07
SNR = 7.53
r:0 c:2 m:0
Total events: 836870
Landau-Gauss fit, trigg.
L MPV = 30.18 ± 0.07
Max = 31.98 ± 0.07
χ²/ndf = 21.88
SNR = 7.79
r:0 c:3 m:0
Total events: 839669
Landau-Gauss fit, trigg.
L MPV = 30.07 ± 0.08
Max = 31.76 ± 0.09
χ²/ndf = 9.51
SNR = 5.52
counts
ADC (arb. units)

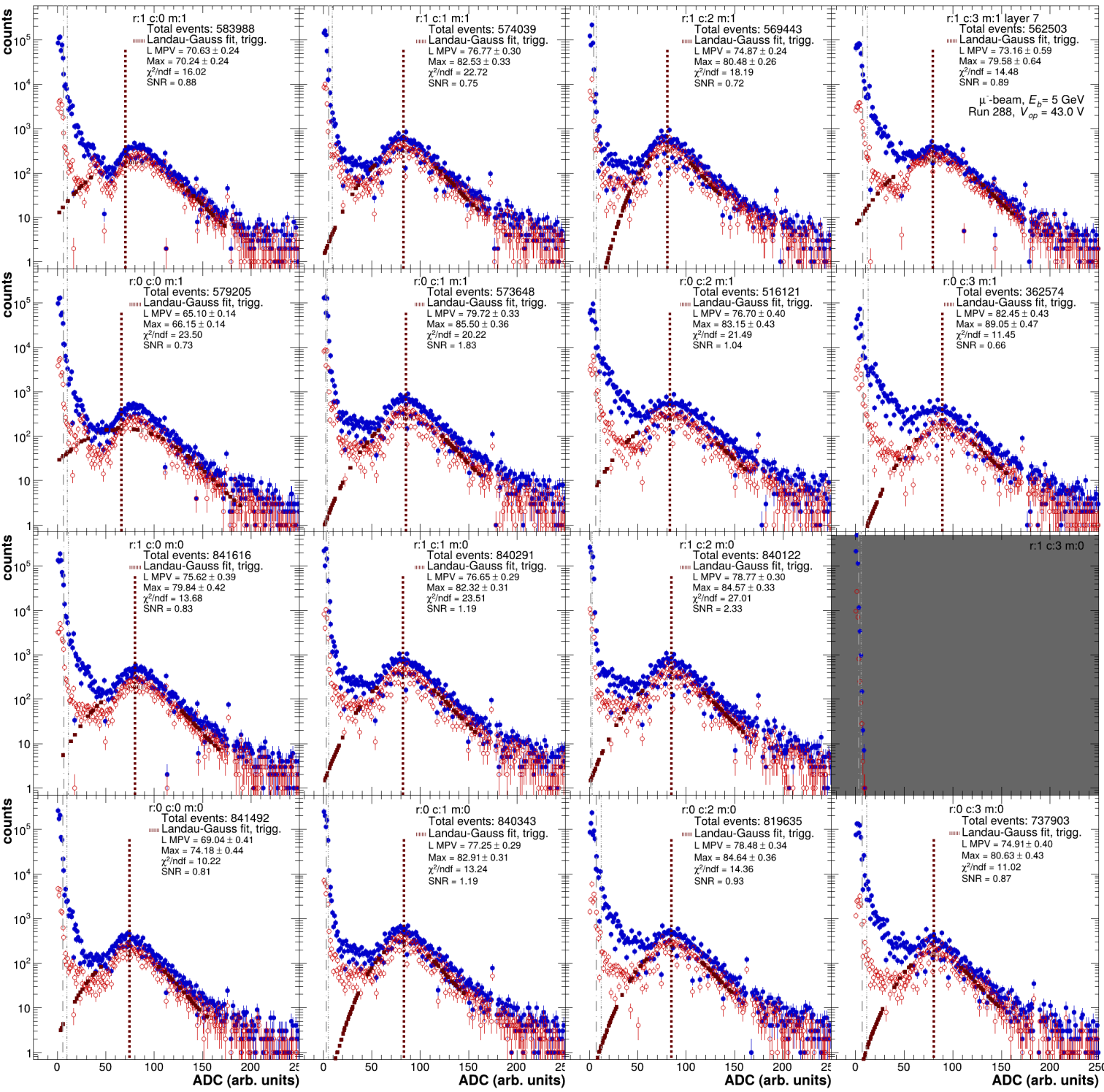

r:1 c:0 m:1
Total events: 519433
Landau-Gauss fit, trigg.
L MPV = 40.83 ± 0.11
Max = 44.41 ± 0.12
χ²/ndf = 19.87
SNR = 5.25
r:1 c:1 m:1
Total events: 575235
Landau-Gauss fit, trigg.
L MPV = 40.26 ± 0.10
Max = 44.74 ± 0.11
χ²/ndf = 25.92
SNR = 7.04
r:1 c:2 m:1
Total events: 512002
Landau-Gauss fit, trigg.
L MPV = 45.46 ± 0.11
Max = 49.91 ± 0.12
χ²/ndf = 32.94
SNR = 6.40
r:1 c:3 m:1 layer 2
Total events: 215521
Landau-Gauss fit, trigg.
L MPV = 41.41 ± 0.11
Max = 45.10 ± 0.12
χ²/ndf = 42.67
SNR = 5.30
μ-beam, E_b = 5 GeV
Run 288, V_op = 43.0 V
r:0 c:0 m:1
Total events: 556815
Landau-Gauss fit, trigg.
L MPV = 40.49 ± 0.12
Max = 44.60 ± 0.13
χ²/ndf = 20.86
SNR = 4.05
r:0 c:1 m:1
Total events: 576446
Landau-Gauss fit, trigg.
L MPV = 39.77 ± 0.14
Max = 45.22 ± 0.15
χ²/ndf = 25.16
SNR = 4.50
r:0 c:2 m:1
Total events: 577400
Landau-Gauss fit, trigg.
L MPV = 37.45 ± 0.13
Max = 42.28 ± 0.14
χ²/ndf = 21.38
SNR = 4.28
r:0 c:3 m:1
Total events: 578449
Landau-Gauss fit, trigg.
L MPV = 43.57 ± 0.15
Max = 47.92 ± 0.16
χ²/ndf = 14.63
SNR = 3.72
r:1 c:0 m:0
Total events: 817598
Landau-Gauss fit, trigg.
L MPV = 42.26 ± 0.12
Max = 45.98 ± 0.13
χ²/ndf = 17.87
SNR = 5.49
r:1 c:1 m:0
Total events: 836521
Landau-Gauss fit, trigg.
L MPV = 35.94 ± 0.08
Max = 38.87 ± 0.08
χ²/ndf = 21.53
SNR = 5.88
r:1 c:2 m:0
Total events: 814025
Landau-Gauss fit, trigg.
L MPV = 40.31 ± 0.09
Max = 43.70 ± 0.10
χ²/ndf = 23.20
SNR = 4.90
r:1 c:3 m:0
Total events: 665713
Landau-Gauss fit, trigg.
L MPV = 39.90 ± 0.10
Max = 43.27 ± 0.11
χ²/ndf = 41.36
SNR = 5.26
r:0 c:0 m:0
Total events: 836986
Landau-Gauss fit, trigg.
L MPV = 41.41 ± 0.14
Max = 45.26 ± 0.15
χ²/ndf = 11.10
SNR = 5.86
r:0 c:1 m:0
Total events: 838143
Landau-Gauss fit, trigg.
L MPV = 39.52 ± 0.10
Max = 43.02 ± 0.11
χ²/ndf = 18.40
SNR = 5.24
r:0 c:2 m:0
Total events: 838557
Landau-Gauss fit, trigg.
L MPV = 36.63 ± 0.11
Max = 40.34 ± 0.12
χ²/ndf = 22.41
SNR = 5.12
r:0 c:3 m:0
Total events: 839256
Landau-Gauss fit, trigg.
L MPV = 38.02 ± 0.12
Max = 40.97 ± 0.13
χ²/ndf = 13.13
SNR = 4.80
counts
ADC (arb. units)

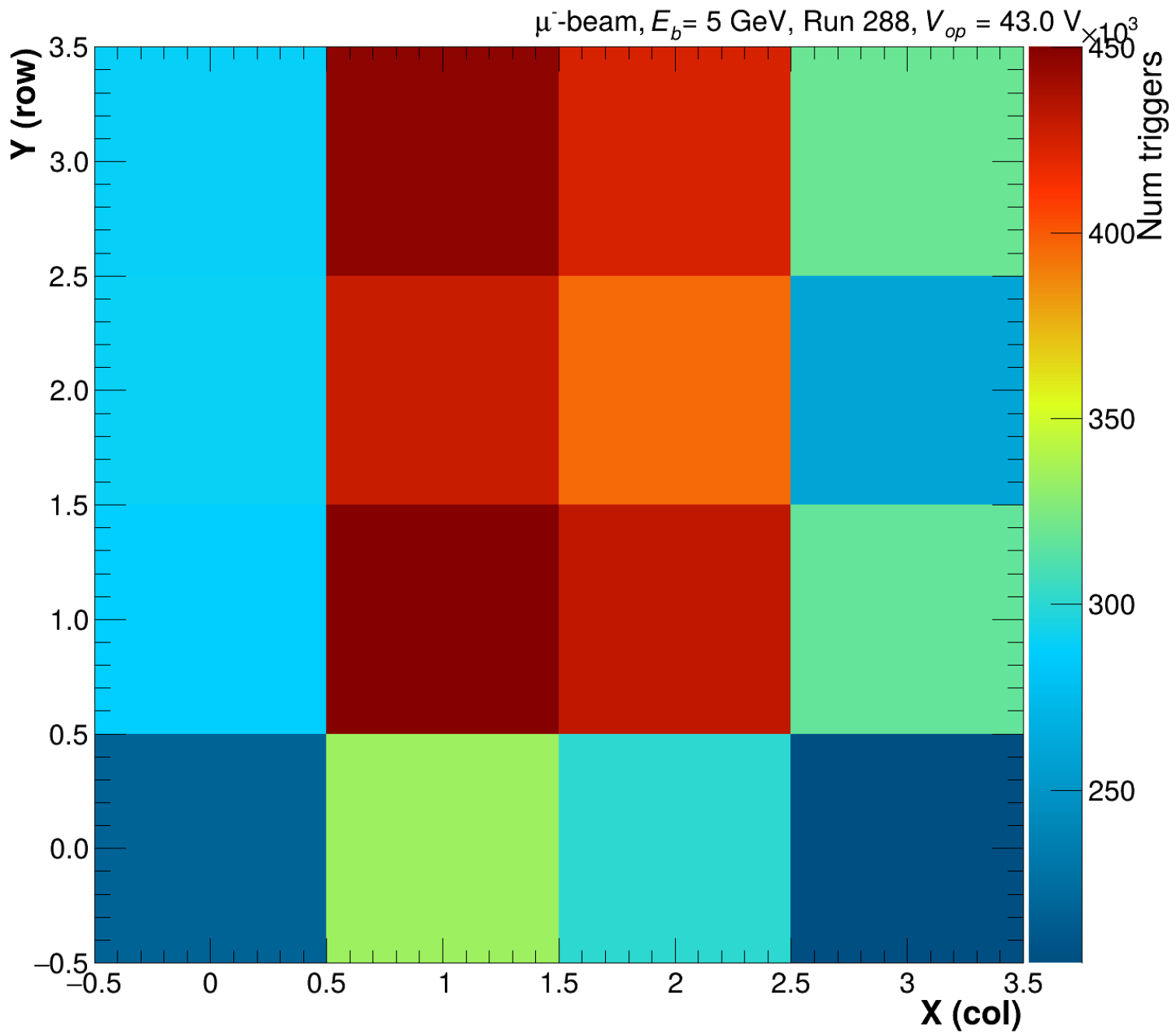

r:1 c:0 m:1
Total events: 569233
Landau-Gauss fit, trigg.
L MPV = 41.06 ± 0.11
Max = 44.44 ± 0.12
χ²/ndf = 24.00
SNR = 10.82
r:1 c:1 m:1
Total events: 301468
Landau-Gauss fit, trigg.
L MPV = 43.58 ± 0.10
Max = 47.73 ± 0.11
χ²/ndf = 48.46
SNR = 3.58
r:1 c:2 m:1
Total events: 575448
Landau-Gauss fit, trigg.
L MPV = 38.81 ± 0.15
Max = 44.18 ± 0.17
χ²/ndf = 42.55
SNR = 3.15
r:1 c:3 m:1 layer 3
Total events: 581225
Landau-Gauss fit, trigg.
L MPV = 42.38 ± 0.11
Max = 45.84 ± 0.12
χ²/ndf = 18.98
SNR = 3.20
μ⁻-beam, E_b = 5 GeV
Run 288, V_op = 43.0 V
r:0 c:0 m:1
Total events: 570929
Landau-Gauss fit, trigg.
L MPV = 42.16 ± 0.12
Max = 45.74 ± 0.13
χ²/ndf = 24.34
SNR = 6.77
r:0 c:1 m:1
Total events: 557598
Landau-Gauss fit, trigg.
L MPV = 43.47 ± 0.11
Max = 47.77 ± 0.12
χ²/ndf = 28.49
SNR = 2.64
r:0 c:2 m:1
Total events: 509943
Landau-Gauss fit, trigg.
L MPV = 40.23 ± 0.11
Max = 44.35 ± 0.13
χ²/ndf = 26.76
SNR = 3.08
r:0 c:3 m:1
Total events: 575810
Landau-Gauss fit, trigg.
L MPV = 39.30 ± 0.13
Max = 42.93 ± 0.14
χ²/ndf = 15.21
SNR = 2.65
r:1 c:0 m:0
Total events: 837928
Landau-Gauss fit, trigg.
L MPV = 42.13 ± 0.12
Max = 45.74 ± 0.13
χ²/ndf = 31.73
SNR = 13.52
r:1 c:1 m:0
Total events: 709399
Landau-Gauss fit, trigg.
L MPV = 40.52 ± 0.09
Max = 43.90 ± 0.10
χ²/ndf = 64.52
SNR = 4.17
r:1 c:2 m:0
Total events: 837602
Landau-Gauss fit, trigg.
L MPV = 39.95 ± 0.10
Max = 43.65 ± 0.11
χ²/ndf = 26.18
SNR = 2.86
r:1 c:3 m:0
Total events: 838069
Landau-Gauss fit, trigg.
L MPV = 43.69 ± 0.12
Max = 47.70 ± 0.13
χ²/ndf = 19.59
SNR = 2.90
r:0 c:0 m:0
Total events: 838834
Landau-Gauss fit, trigg.
L MPV = 40.21 ± 0.12
Max = 43.23 ± 0.13
χ²/ndf = 20.85
SNR = 7.75
r:0 c:1 m:0
Total events: 837106
Landau-Gauss fit, trigg.
L MPV = 42.96 ± 0.11
Max = 46.85 ± 0.12
χ²/ndf = 17.32
SNR = 3.27
r:0 c:2 m:0
Total events: 817275
Landau-Gauss fit, trigg.
L MPV = 41.63 ± 0.12
Max = 45.28 ± 0.13
χ²/ndf = 21.00
SNR = 3.57
r:0 c:3 m:0
Total events: 839500
Landau-Gauss fit, trigg.
L MPV = 43.06 ± 0.15
Max = 47.02 ± 0.17
χ²/ndf = 10.29
SNR = 2.59
counts
ADC (arb. units)

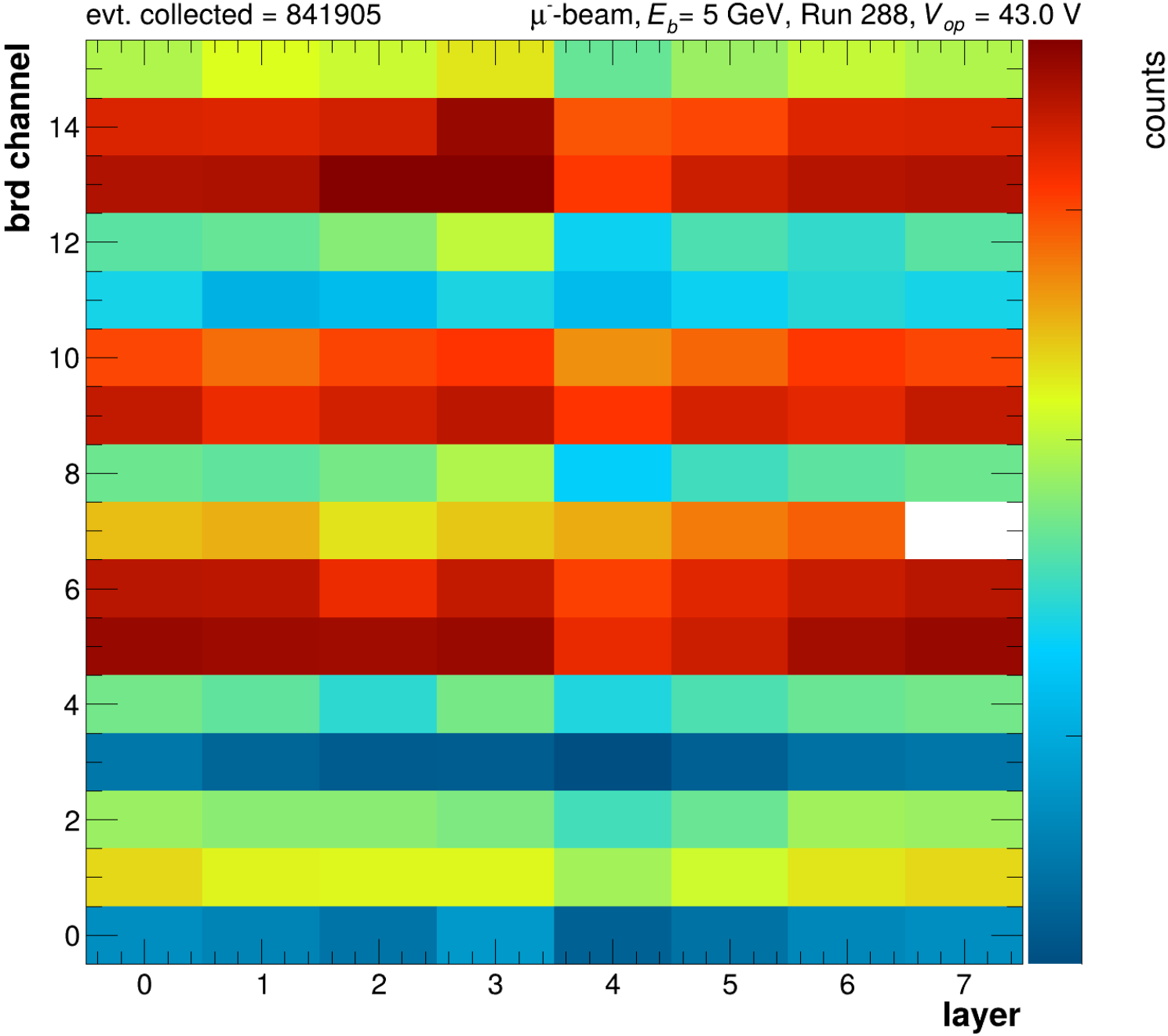

r:1 c:0 m:1
Total events: 583165
Landau-Gauss fit, trigg.
L MPV = 51.72 ± 0.16
Max = 56.33 ± 0.18
χ²/ndf = 20.24
SNR = 2.59
r:1 c:1 m:1
Total events: 580665
Landau-Gauss fit, trigg.
L MPV = 57.14 ± 0.16
Max = 62.39 ± 0.17
χ²/ndf = 28.25
SNR = 2.45
r:1 c:2 m:1
Total events: 578882
Landau-Gauss fit, trigg.
L MPV = 50.88 ± 0.14
Max = 55.73 ± 0.16
χ²/ndf = 20.10
SNR = 2.25
r:1 c:3 m:1 layer 4
Total events: 565462
Landau-Gauss fit, trigg.
L MPV = 59.80 ± 0.19
Max = 65.48 ± 0.21
χ²/ndf = 24.40
SNR = 2.46
μ⁻-beam, E_b = 5 GeV
Run 288, V_op = 43.0 V
r:0 c:0 m:1
Total events: 568111
Landau-Gauss fit, trigg.
L MPV = 61.27 ± 0.23
Max = 66.74 ± 0.25
χ²/ndf = 19.17
SNR = 2.17
r:0 c:1 m:1
Total events: 582055
Landau-Gauss fit, trigg.
L MPV = 51.65 ± 0.16
Max = 56.46 ± 0.17
χ²/ndf = 25.67
SNR = 2.02
r:0 c:2 m:1
Total events: 564156
Landau-Gauss fit, trigg.
L MPV = 60.71 ± 0.20
Max = 66.25 ± 0.21
χ²/ndf = 27.87
SNR = 1.98
r:0 c:3 m:1
Total events: 554832
Landau-Gauss fit, trigg.
L MPV = 60.11 ± 0.17
Max = 64.39 ± 0.18
χ²/ndf = 8.91
SNR = 1.84
r:1 c:0 m:0
Total events: 840474
Landau-Gauss fit, trigg.
L MPV = 56.10 ± 0.17
Max = 61.04 ± 0.19
χ²/ndf = 17.22
SNR = 2.26
r:1 c:1 m:0
Total events: 839424
Landau-Gauss fit, trigg.
L MPV = 55.87 ± 0.14
Max = 60.75 ± 0.15
χ²/ndf = 25.20
SNR = 2.40
r:1 c:2 m:0
Total events: 839508
Landau-Gauss fit, trigg.
L MPV = 53.51 ± 0.14
Max = 58.21 ± 0.15
χ²/ndf = 24.09
SNR = 2.28
r:1 c:3 m:0
Total events: 838233
Landau-Gauss fit, trigg.
L MPV = 56.05 ± 0.16
Max = 61.47 ± 0.18
χ²/ndf = 27.28
SNR = 2.18
r:0 c:0 m:0
Total events: 839541
Landau-Gauss fit, trigg.
L MPV = 56.00 ± 0.18
Max = 60.44 ± 0.19
χ²/ndf = 12.57
SNR = 2.32
r:0 c:1 m:0
Total events: 840228
Landau-Gauss fit, trigg.
L MPV = 58.67 ± 0.18
Max = 63.65 ± 0.19
χ²/ndf = 19.23
SNR = 2.48
r:0 c:2 m:0
Total events: 838555
Landau-Gauss fit, trigg.
L MPV = 57.57 ± 0.18
Max = 62.63 ± 0.20
χ²/ndf = 21.12
SNR = 2.55
r:0 c:3 m:0
Total events: 837746
Landau-Gauss fit, trigg.
L MPV = 57.51 ± 0.16
Max = 61.29 ± 0.18
χ²/ndf = 7.84
SNR = 2.61
counts
ADC (arb. units)

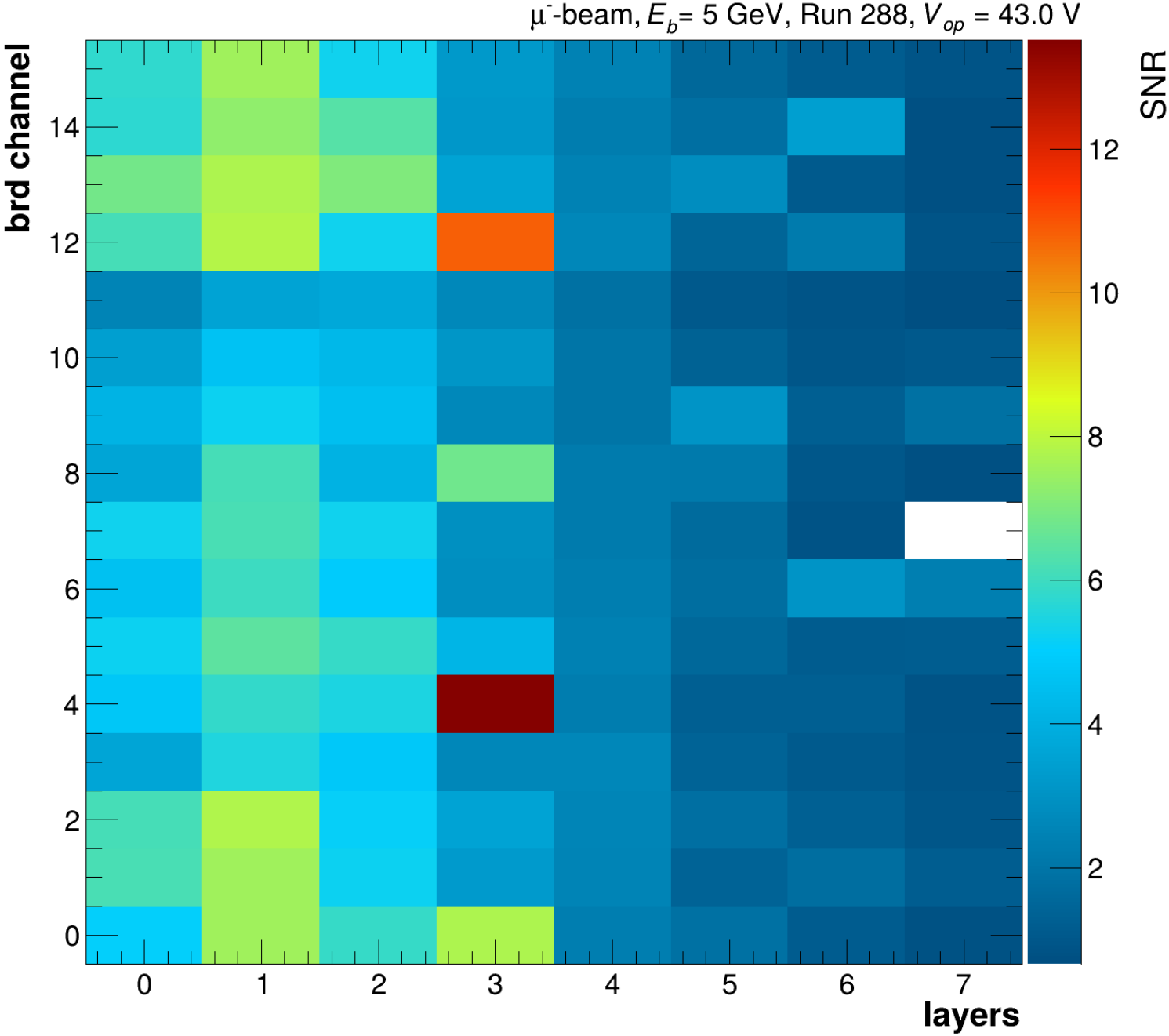

r:1 c:0 m:1
Total events: 573727
Landau-Gauss fit, trigg.
L MPV = 57.07 ± 0.14
Max = 60.57 ± 0.15
χ²/ndf = 13.56
SNR = 1.45
r:1 c:1 m:1
Total events: 574679
Landau-Gauss fit, trigg.
L MPV = 56.39 ± 0.11
Max = 60.00 ± 0.12
χ²/ndf = 17.75
SNR = 2.78
r:1 c:2 m:1
Total events: 516025
Landau-Gauss fit, trigg.
L MPV = 66.45 ± 0.16
Max = 71.40 ± 0.17
χ²/ndf = 19.22
SNR = 1.78
r:1 c:3 m:1 layer 5
Total events: 565829
Landau-Gauss fit, trigg.
L MPV = 60.79 ± 0.22
Max = 65.96 ± 0.24
χ²/ndf = 21.23
SNR = 1.56
μ⁻-beam, E_b = 5 GeV
Run 288, V_op = 43.0 V
r:0 c:0 m:1
Total events: 584283
Landau-Gauss fit, trigg.
L MPV = 60.47 ± 0.19
Max = 65.03 ± 0.20
χ²/ndf = 10.12
SNR = 2.15
r:0 c:1 m:1
Total events: 571299
Landau-Gauss fit, trigg.
L MPV = 59.58 ± 0.13
Max = 63.59 ± 0.14
χ²/ndf = 20.76
SNR = 3.03
r:0 c:2 m:1
Total events: 563531
Landau-Gauss fit, trigg.
L MPV = 66.10 ± 0.26
Max = 71.74 ± 0.28
χ²/ndf = 29.85
SNR = 1.35
r:0 c:3 m:1
Total events: 579287
Landau-Gauss fit, trigg.
L MPV = 63.51 ± 0.21
Max = 68.28 ± 0.23
χ²/ndf = 6.97
SNR = 1.04
r:1 c:0 m:0
Total events: 839896
Landau-Gauss fit, trigg.
L MPV = 63.68 ± 0.18
Max = 68.45 ± 0.20
χ²/ndf = 18.85
SNR = 1.25
r:1 c:1 m:0
Total events: 839446
Landau-Gauss fit, trigg.
L MPV = 68.10 ± 0.16
Max = 73.16 ± 0.17
χ²/ndf = 26.29
SNR = 1.52
r:1 c:2 m:0
Total events: 819012
Landau-Gauss fit, trigg.
L MPV = 55.63 ± 0.14
Max = 60.00 ± 0.15
χ²/ndf = 21.58
SNR = 1.77
r:1 c:3 m:0
Total events: 839179
Landau-Gauss fit, trigg.
L MPV = 61.99 ± 0.20
Max = 67.46 ± 0.22
χ²/ndf = 20.41
SNR = 1.66
r:0 c:0 m:0
Total events: 841374
Landau-Gauss fit, trigg.
L MPV = 64.89 ± 0.26
Max = 70.07 ± 0.28
χ²/ndf = 12.66
SNR = 1.85
r:0 c:1 m:0
Total events: 839796
Landau-Gauss fit, trigg.
L MPV = 64.69 ± 0.17
Max = 69.45 ± 0.19
χ²/ndf = 27.88
SNR = 1.41
r:0 c:2 m:0
Total events: 839217
Landau-Gauss fit, trigg.
L MPV = 64.14 ± 0.22
Max = 69.23 ± 0.24
χ²/ndf = 16.11
SNR = 1.80
r:0 c:3 m:0
Total events: 841151
Landau-Gauss fit, trigg.
L MPV = 62.77 ± 0.23
Max = 67.44 ± 0.24
χ²/ndf = 9.53
SNR = 1.37
counts
ADC (arb. units)

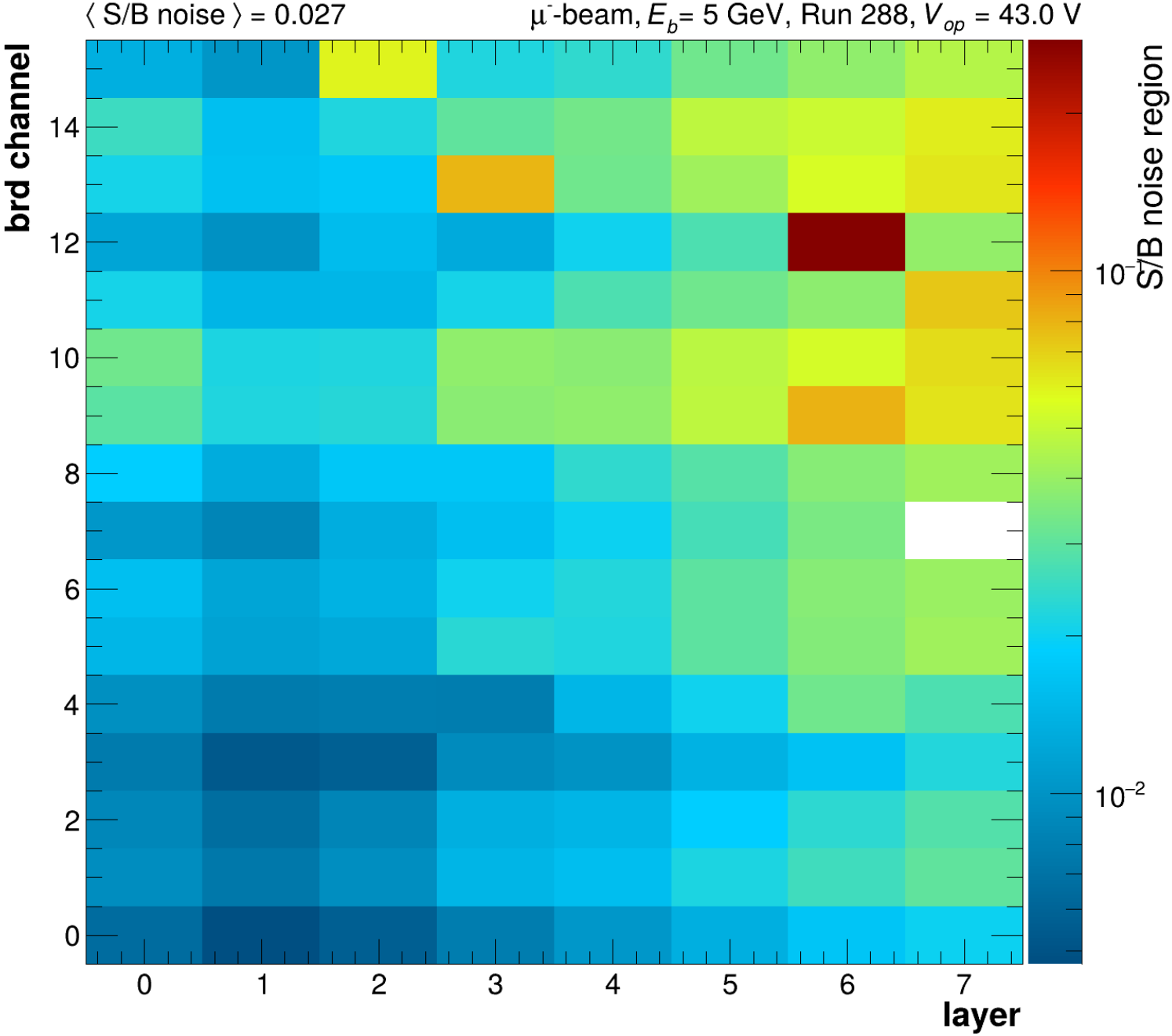

r:1 c:0 m:1
Total events: 120489
Landau-Gauss fit, trigg.
L MPV = 75.55 ± 0.34
Max = 81.07 ± 0.36
χ²/ndf = 22.72
SNR = 2.21
r:1 c:1 m:1
Total events: 577485
Landau-Gauss fit, trigg.
L MPV = 73.01 ± 0.19
Max = 78.38 ± 0.20
χ²/ndf = 13.88
SNR = 1.08
r:1 c:2 m:1
Total events: 569224
Landau-Gauss fit, trigg.
L MPV = 66.27 ± 0.20
Max = 71.38 ± 0.22
χ²/ndf = 27.02
SNR = 3.42
r:1 c:3 m:1 layer 6
Total events: 578081
Landau-Gauss fit, trigg.
L MPV = 75.34 ± 0.25
Max = 80.98 ± 0.27
χ²/ndf = 8.65
SNR = 1.15
μ⁻-beam, E_b = 5 GeV
Run 288, V_op = 43.0 V
r:0 c:0 m:1
Total events: 583709
Landau-Gauss fit, trigg.
L MPV = 66.62 ± 0.46
Max = 70.65 ± 0.48
χ²/ndf = 12.84
SNR = 1.00
r:0 c:1 m:1
Total events: 432520
Landau-Gauss fit, trigg.
L MPV = 76.90 ± 0.24
Max = 82.67 ± 0.26
χ²/ndf = 21.77
SNR = 1.22
r:0 c:2 m:1
Total events: 574608
Landau-Gauss fit, trigg.
L MPV = 69.55 ± 0.21
Max = 74.61 ± 0.22
χ²/ndf = 11.88
SNR = 0.93
r:0 c:3 m:1
Total events: 578023
Landau-Gauss fit, trigg.
L MPV = 70.07 ± 0.26
Max = 75.20 ± 0.28
χ²/ndf = 6.38
SNR = 0.88
r:1 c:0 m:0
Total events: 625149
Landau-Gauss fit, trigg.
L MPV = 66.26 ± 0.26
Max = 71.56 ± 0.28
χ²/ndf = 21.99
SNR = 1.25
r:1 c:1 m:0
Total events: 840866
Landau-Gauss fit, trigg.
L MPV = 71.01 ± 0.17
Max = 76.20 ± 0.19
χ²/ndf = 18.52
SNR = 1.11
r:1 c:2 m:0
Total events: 839828
Landau-Gauss fit, trigg.
L MPV = 70.09 ± 0.20
Max = 75.54 ± 0.21
χ²/ndf = 32.14
SNR = 3.05
r:1 c:3 m:0
Total events: 841152
Landau-Gauss fit, trigg.
L MPV = 64.56 ± 0.25
Max = 69.82 ± 0.27
χ²/ndf = 17.57
SNR = 0.84
r:0 c:0 m:0
Total events: 841552
Landau-Gauss fit, trigg.
L MPV = 68.40 ± 0.27
Max = 73.58 ± 0.30
χ²/ndf = 8.22
SNR = 1.14
r:0 c:1 m:0
Total events: 773739
Landau-Gauss fit, trigg.
L MPV = 65.16 ± 0.23
Max = 70.35 ± 0.25
χ²/ndf = 27.90
SNR = 1.76
r:0 c:2 m:0
Total events: 840298
Landau-Gauss fit, trigg.
L MPV = 69.58 ± 0.21
Max = 74.56 ± 0.23
χ²/ndf = 10.95
SNR = 1.28
r:0 c:3 m:0
Total events: 841247
Landau-Gauss fit, trigg.
L MPV = 70.60 ± 0.31
Max = 76.08 ± 0.34
χ²/ndf = 10.01
SNR = 1.08
counts
ADC (arb. units)

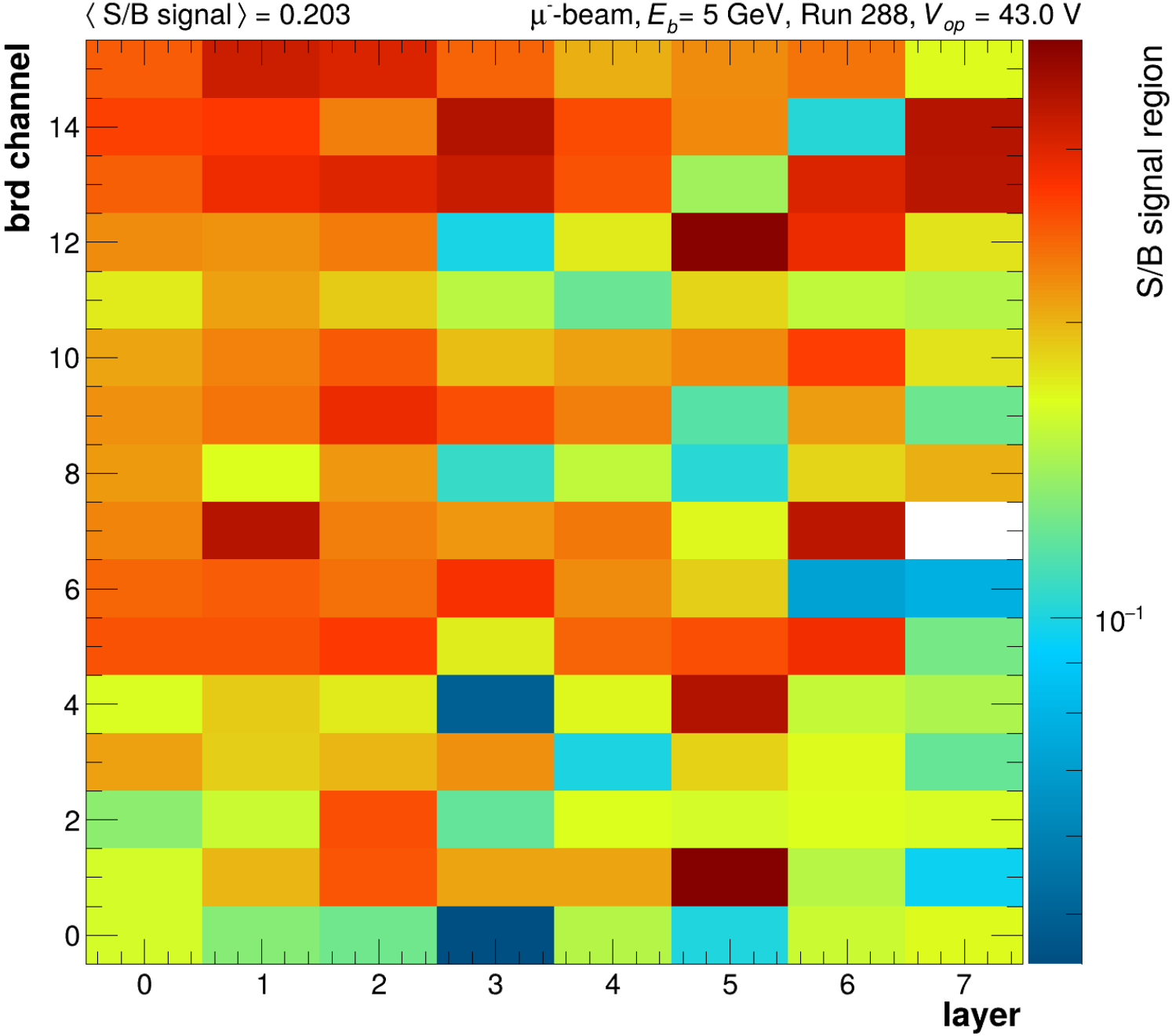

r:1 c:0 m:1
Total events: 583988
Landau-Gauss fit, trigg.
L MPV = 70.63 ± 0.24
Max = 70.24 ± 0.24
χ²/ndf = 16.02
SNR = 0.88
r:1 c:1 m:1
Total events: 574039
Landau-Gauss fit, trigg.
L MPV = 76.77 ± 0.30
Max = 82.53 ± 0.33
χ²/ndf = 22.72
SNR = 0.75
r:1 c:2 m:1
Total events: 569443
Landau-Gauss fit, trigg.
L MPV = 74.87 ± 0.24
Max = 80.48 ± 0.26
χ²/ndf = 18.19
SNR = 0.72
r:1 c:3 m:1 layer 7
Total events: 562503
Landau-Gauss fit, trigg.
L MPV = 73.16 ± 0.59
Max = 79.58 ± 0.64
χ²/ndf = 14.48
SNR = 0.89
μ⁻-beam, E_b = 5 GeV
Run 288, V_op = 43.0 V
r:0 c:0 m:1
Total events: 579205
Landau-Gauss fit, trigg.
L MPV = 65.10 ± 0.14
Max = 66.15 ± 0.14
χ²/ndf = 23.50
SNR = 0.73
r:0 c:1 m:1
Total events: 573648
Landau-Gauss fit, trigg.
L MPV = 79.72 ± 0.33
Max = 85.50 ± 0.36
χ²/ndf = 20.22
SNR = 1.83
r:0 c:2 m:1
Total events: 516121
Landau-Gauss fit, trigg.
L MPV = 76.70 ± 0.40
Max = 83.15 ± 0.43
χ²/ndf = 21.49
SNR = 1.04
r:0 c:3 m:1
Total events: 362574
Landau-Gauss fit, trigg.
L MPV = 82.45 ± 0.43
Max = 89.05 ± 0.47
χ²/ndf = 11.45
SNR = 0.66
r:1 c:0 m:0
Total events: 841616
Landau-Gauss fit, trigg.
L MPV = 75.62 ± 0.39
Max = 79.84 ± 0.42
χ²/ndf = 13.68
SNR = 0.83
r:1 c:1 m:0
Total events: 840291
Landau-Gauss fit, trigg.
L MPV = 76.65 ± 0.29
Max = 82.32 ± 0.31
χ²/ndf = 23.51
SNR = 1.19
r:1 c:2 m:0
Total events: 840122
Landau-Gauss fit, trigg.
L MPV = 78.77 ± 0.30
Max = 84.57 ± 0.33
χ²/ndf = 27.01
SNR = 2.33
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 841492
Landau-Gauss fit, trigg.
L MPV = 69.04 ± 0.41
Max = 74.18 ± 0.44
χ²/ndf = 10.22
SNR = 0.81
r:0 c:1 m:0
Total events: 840343
Landau-Gauss fit, trigg.
L MPV = 77.25 ± 0.29
Max = 82.91 ± 0.31
χ²/ndf = 13.24
SNR = 1.19
r:0 c:2 m:0
Total events: 819635
Landau-Gauss fit, trigg.
L MPV = 78.48 ± 0.34
Max = 84.64 ± 0.36
χ²/ndf = 14.36
SNR = 0.93
r:0 c:3 m:0
Total events: 737903
Landau-Gauss fit, trigg.
L MPV = 74.91 ± 0.40
Max = 80.63 ± 0.43
χ²/ndf = 11.02
SNR = 0.87
counts
ADC (arb. units)

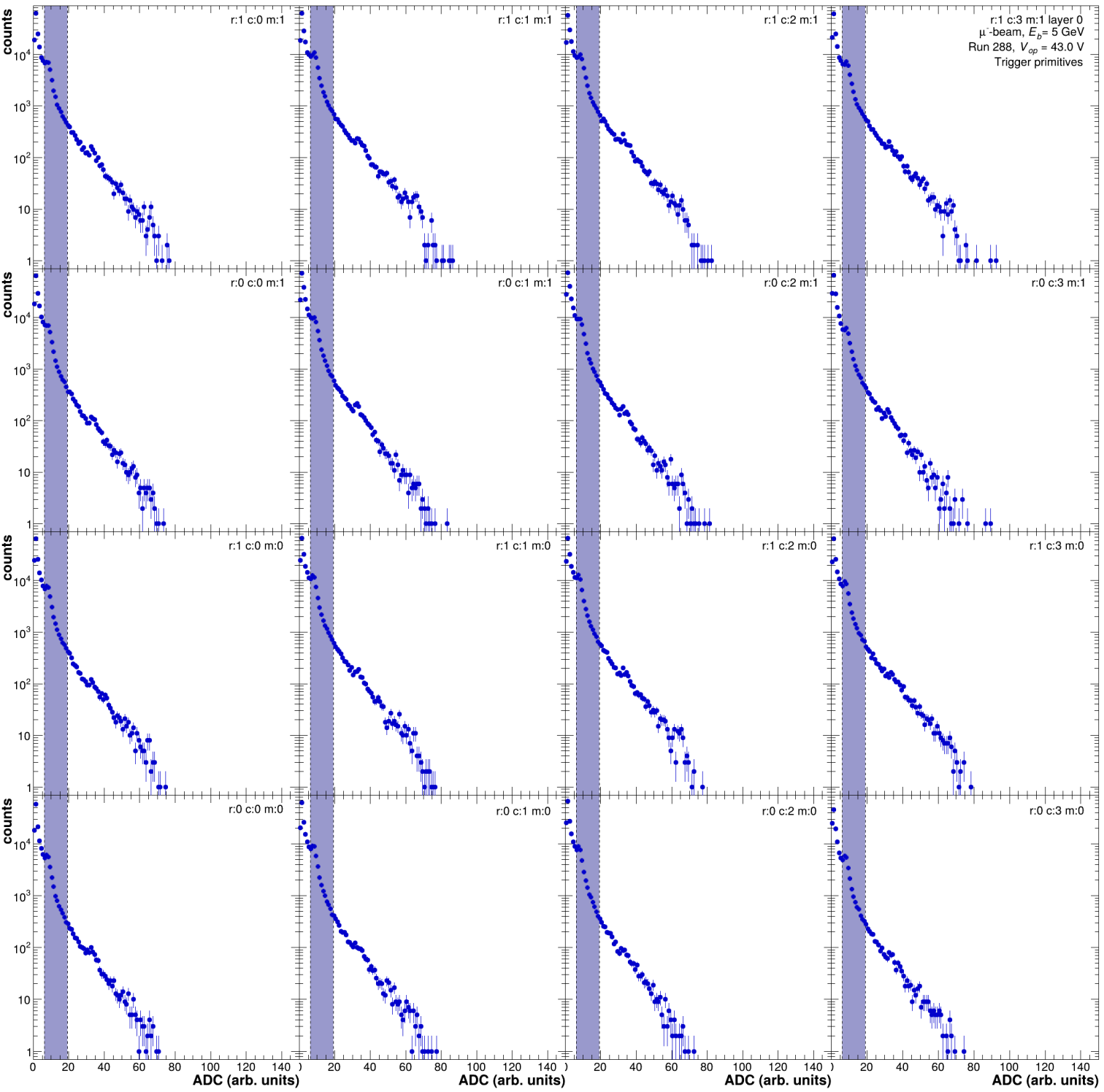

μ⁻-beam, E_b= 5 GeV, Run 288, V_{op} = 43.0 V
Y (row)
X (col)
Num triggers
×10³
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−0.5
0
0.5
1
1.5
2
2.5
3
3.5
450
400
350
300
250

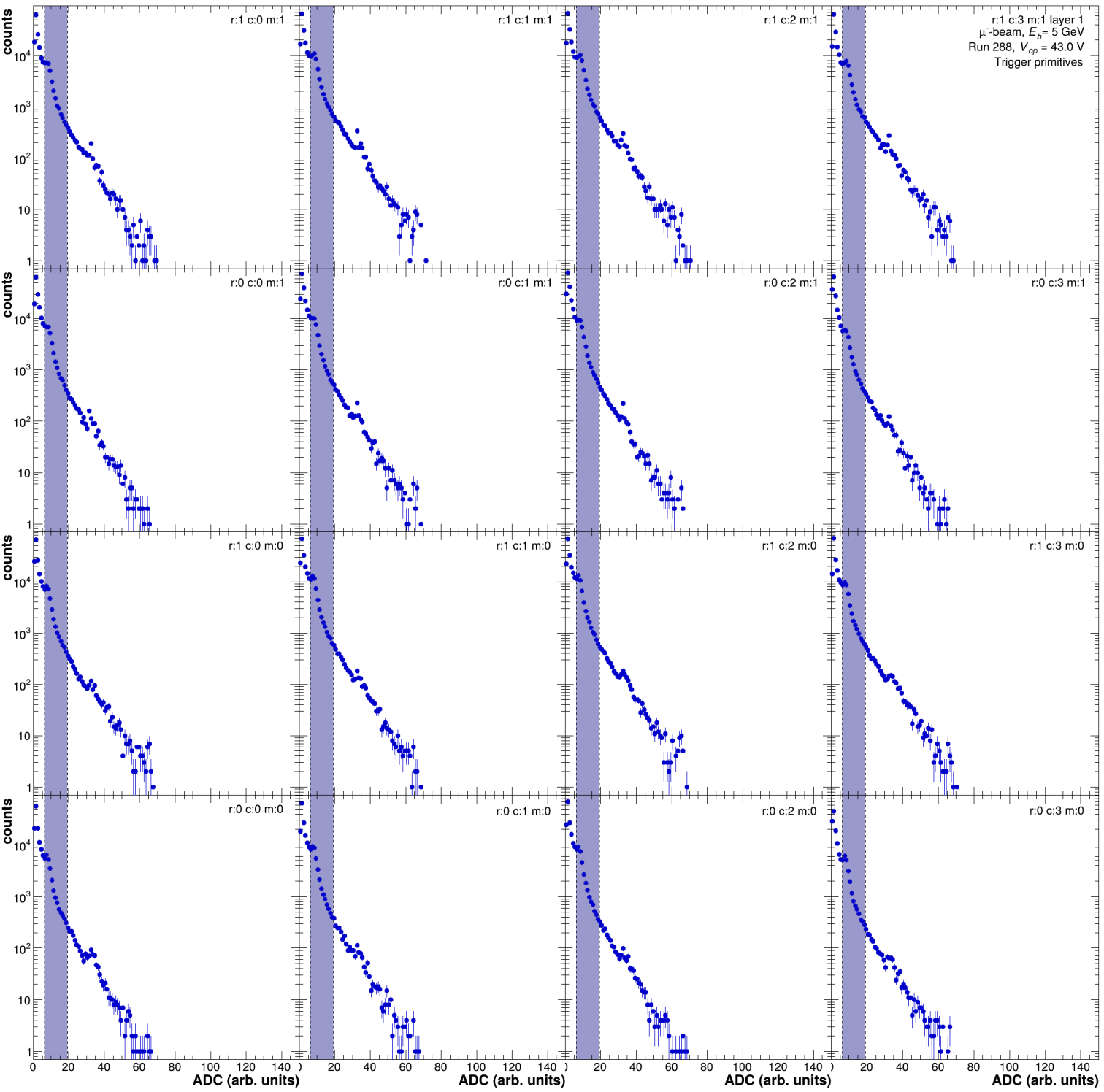

evt. collected = 841905
μ⁻-beam, E_b= 5 GeV, Run 288, V_{op} = 43.0 V
brd channel
14
12
10
8
6
4
2
0
0
1
2
3
4
5
6
7
layer
counts

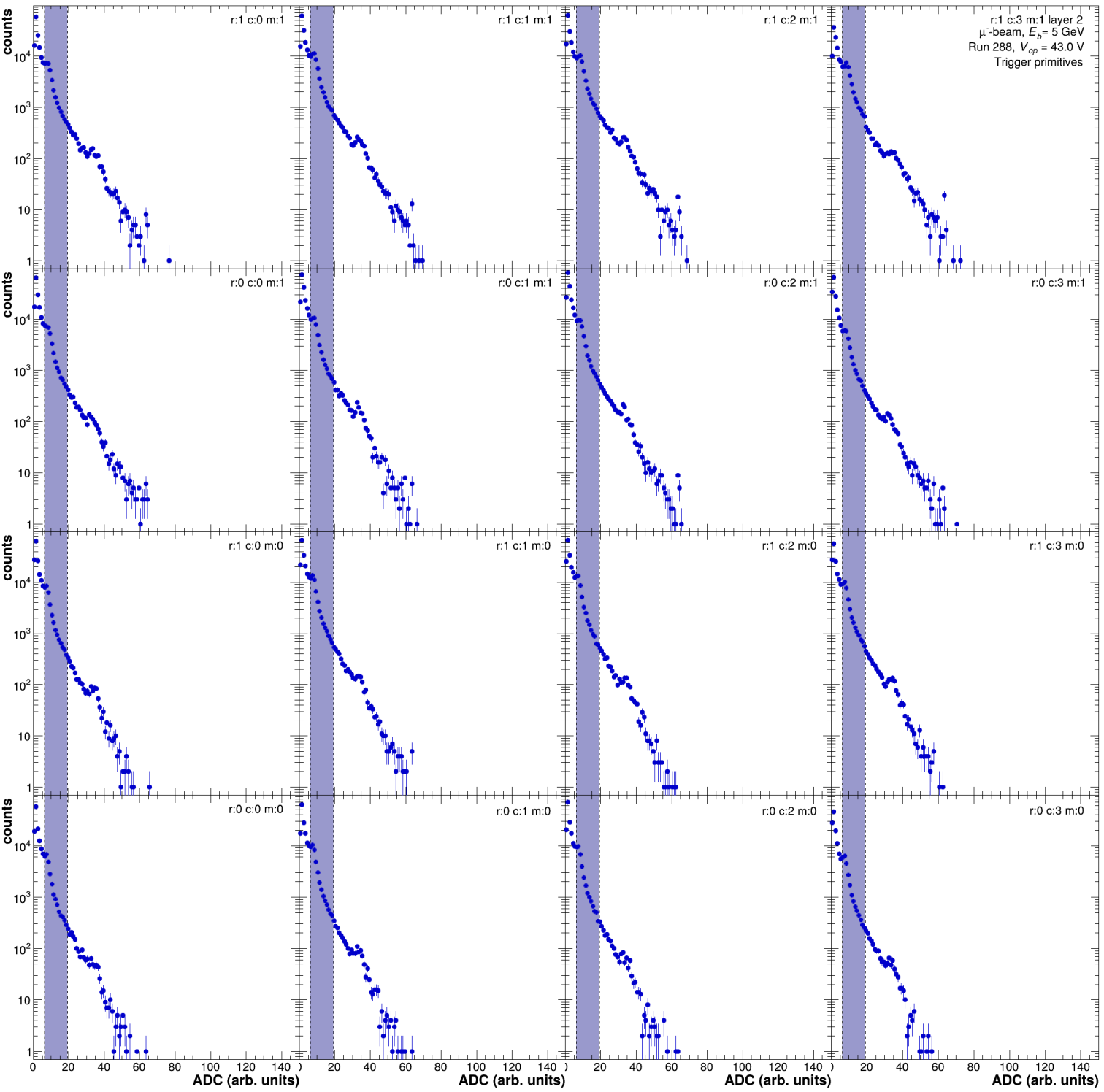

μ⁻-beam, E_b= 5 GeV, Run 288, V_{op} = 43.0 V
brd channel
layers
SNR
0
1
2
3
4
5
6
7
0
2
4
6
8
10
12
14
2
4
6
8
10
12

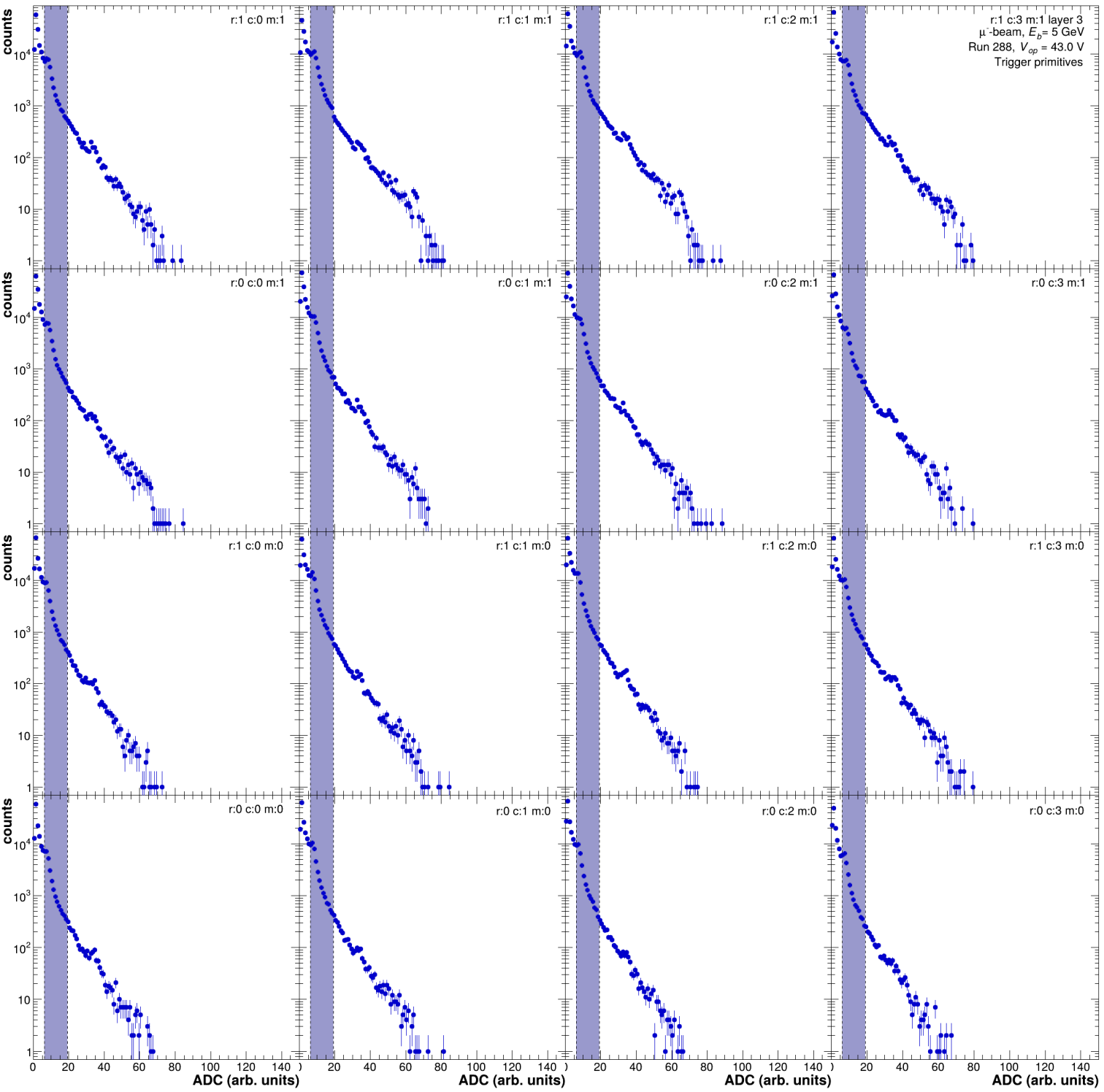

⟨ S/B noise ⟩ = 0.027
μ⁻-beam, E_b= 5 GeV, Run 288, V_{op} = 43.0 V
brd channel
layer
S/B noise region
10^{-1}
10^{-2}
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7

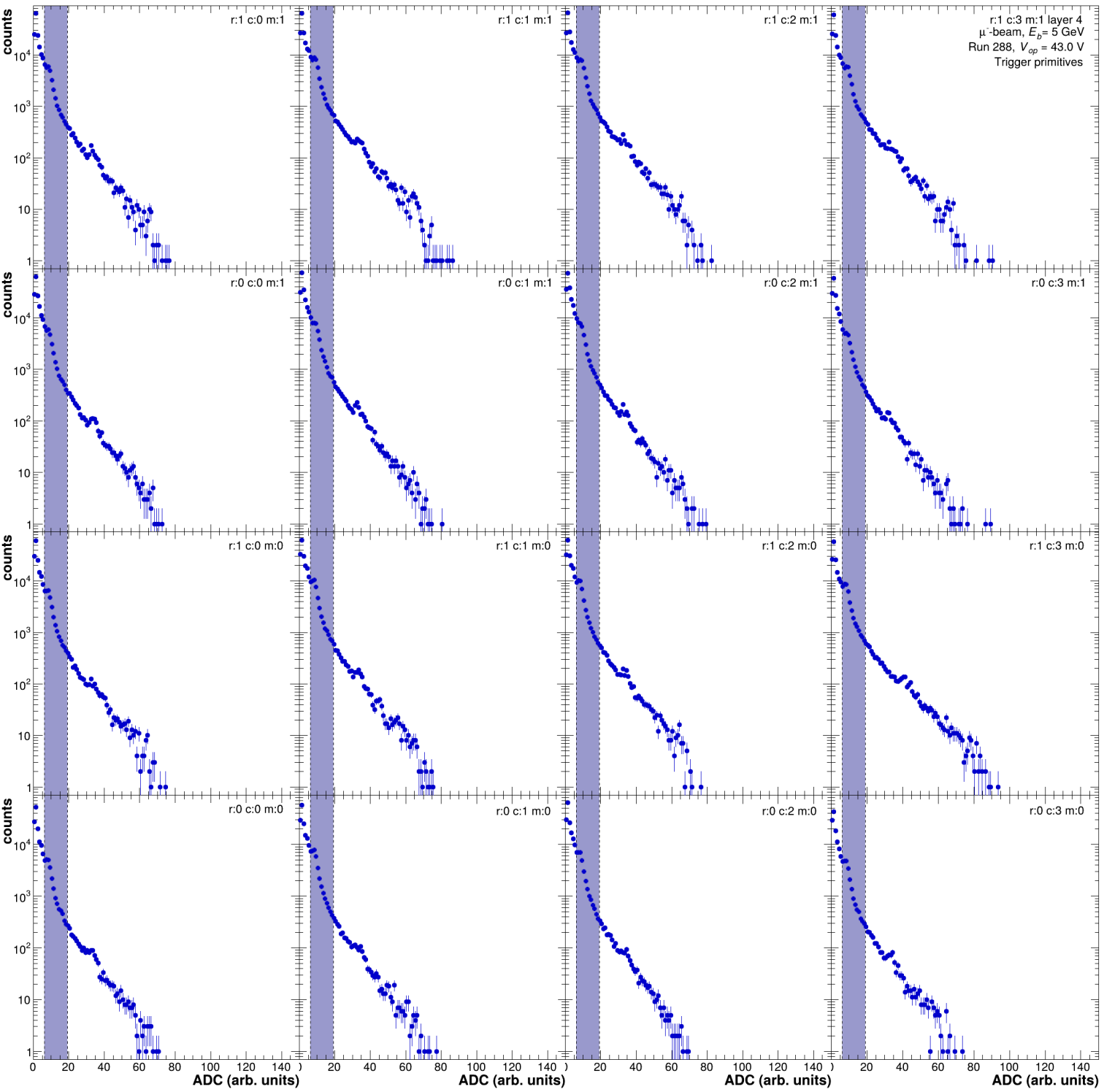

⟨ S/B signal ⟩ = 0.203
μ⁻-beam, E_b= 5 GeV, Run 288, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
S/B signal region
10^{-1}

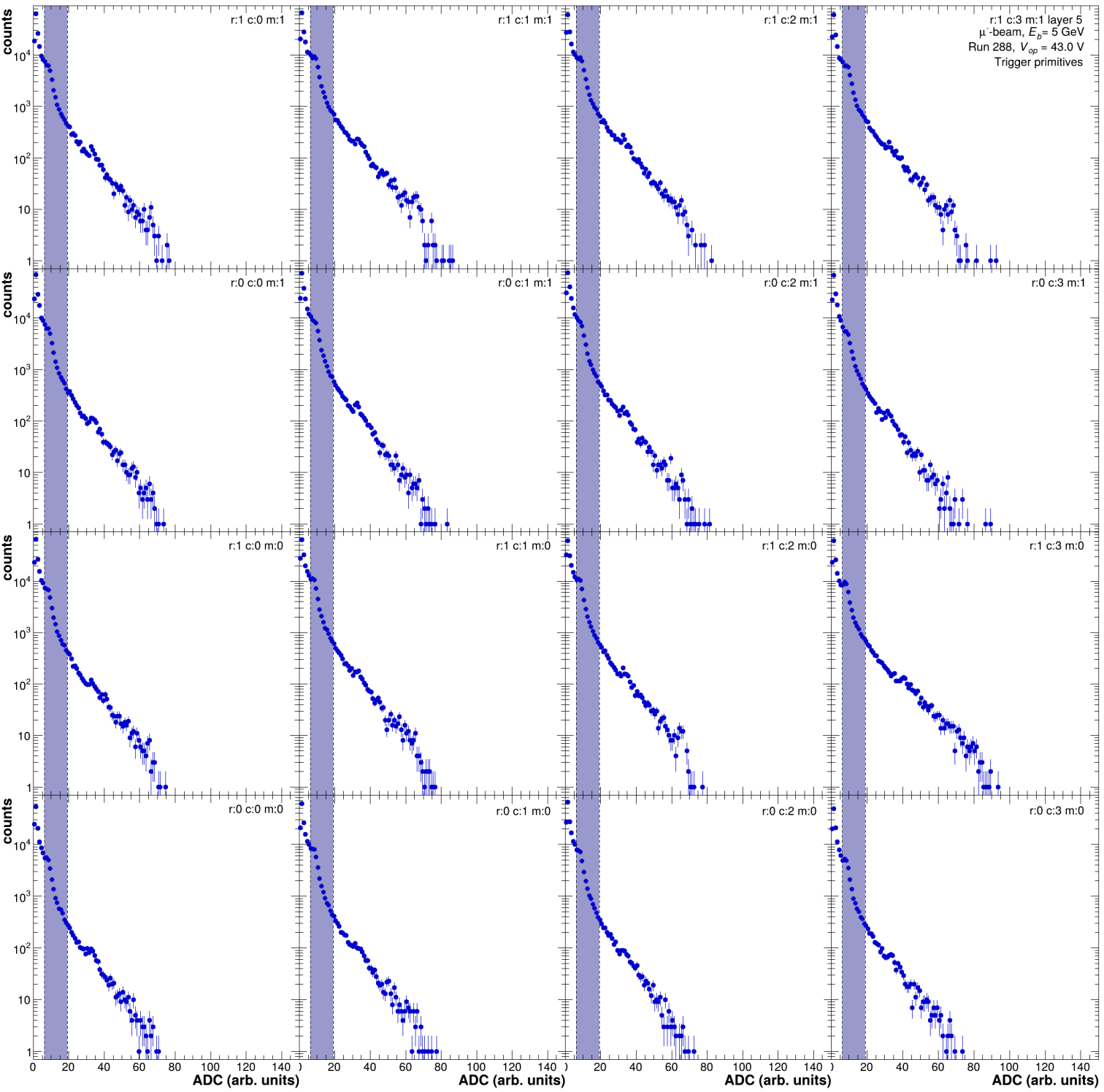

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 0
μ⁻-beam, E_b= 5 GeV
Run 288, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

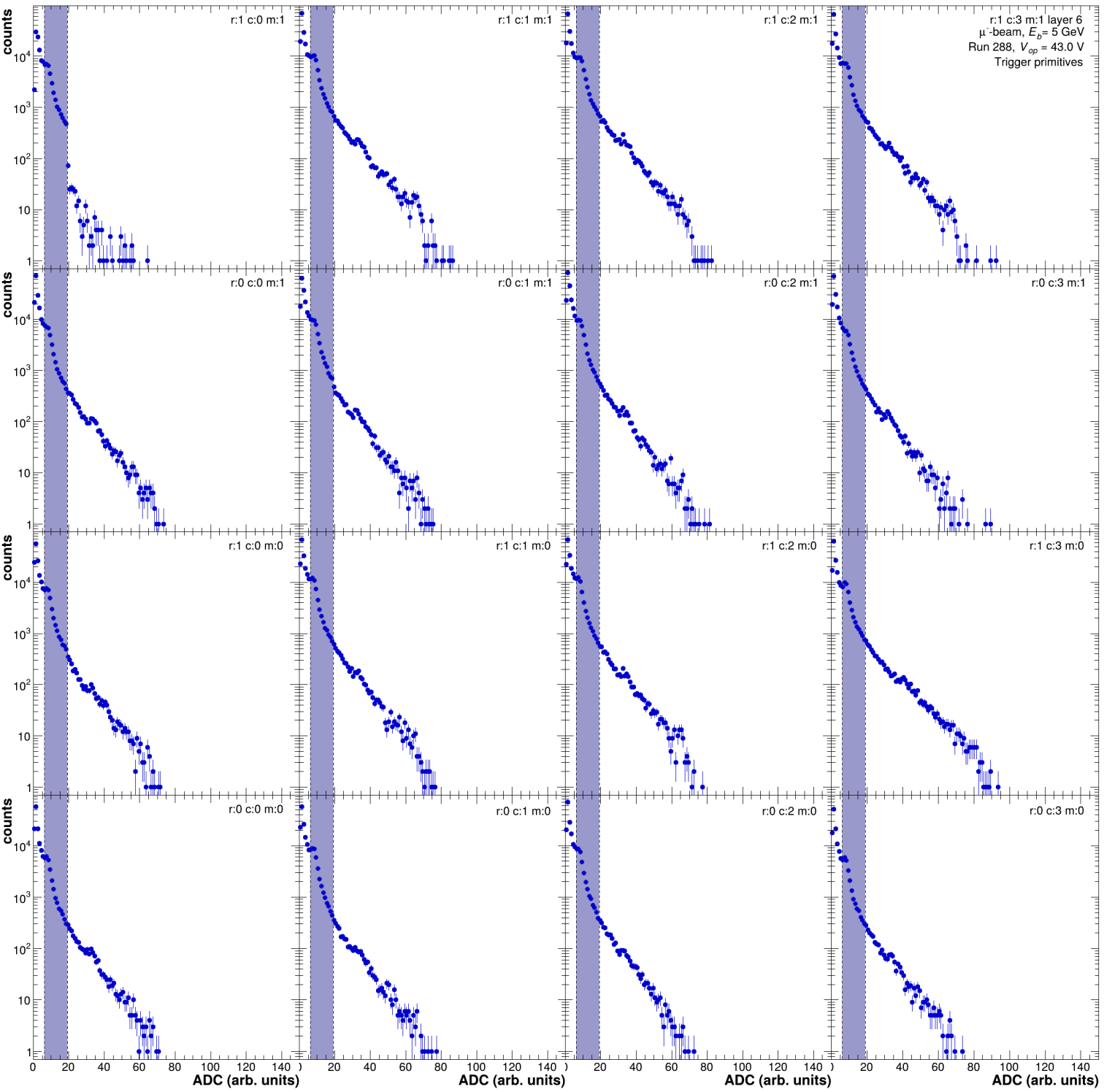

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 1
μ⁻-beam, E_b= 5 GeV
Run 288, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

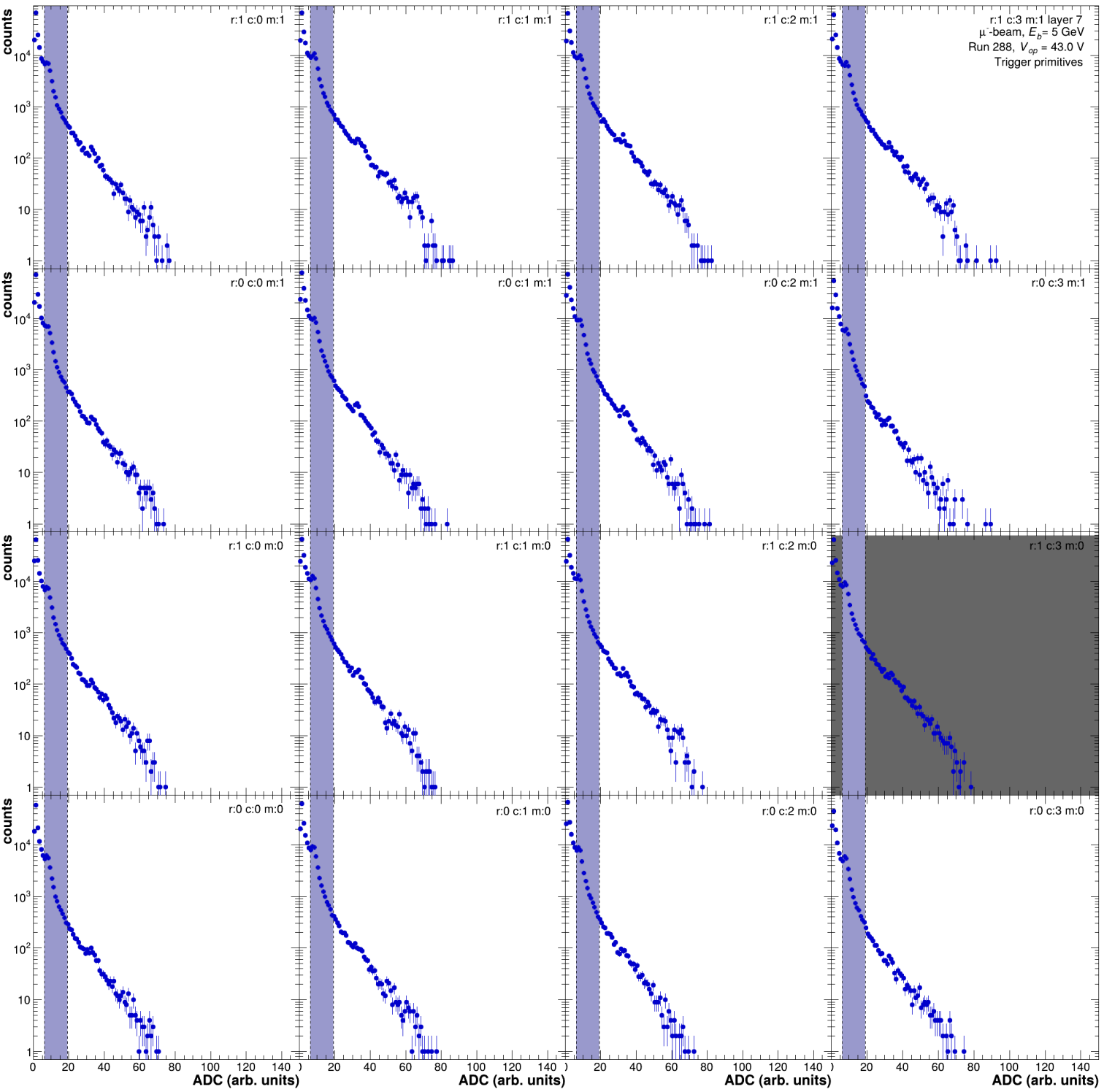

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 2
μ⁻-beam, E_b= 5 GeV
Run 288, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 3
μ⁻-beam, E_b= 5 GeV
Run 288, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 4
μ⁻-beam, E_b= 5 GeV
Run 288, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 5
μ⁻-beam, E_b= 5 GeV
Run 288, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 6
μ⁻-beam, E_b= 5 GeV
Run 288, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 7
μ⁻-beam, E_b= 5 GeV
Run 288, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)